

Product Requirement Document for Glowworm Music Studio

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Change History

Date	Change	Done By
3/11/26	Create doc, add outline, update timeline	Henny
3/25/26	Update Section 1, change submit date, add sprint 1 reflections and outline for other sprint reflections to Section 10, add Trello link to Section 8 and update dates	Henny
3/30/26	Update Sections 2, 3, 8, 10	Henny
4/6/26	Add user persona map to section 3. Update sections 4, 5, 6, 7, 8, 10	Henny
4/13/26	Modify test 1 from section 5. Add low-fid prototype to section 7. Update sections 8 and 10	Henny
4/20/26	Update sections 8-10. Add Wireframe prototype to 7, update 4 to add "REQUIRED" or "OPTIONAL" to each ranking	Henny
4/27/26	Update sections 5, 8-10. Fix the Overview to reflect new changes to PWA	Henny
5/1/26	Update sections 5, 8-10	Henny
5/6/26	Add rest of design methods from first three sprints to section 7. Update sections 8, 10. General SPAG fixes	Henny

I. Overview

Casual and aspiring musicians or other creators often lack intuitive, easily accessible tools for creating, recording, and sharing music in an interactive way. That is, musicians and creators need a tool that allows them to easily experiment with various familiar and unfamiliar instruments, write sheet music or scores without prior knowledge, and easily combine scores to create a hypothetical song that can be shared with other users. To meet user needs, we propose a design for a progressive web application (PWA) where users can play different instruments using their keyboard. Specifically, users can switch between three modes, test mode (Studio)—where the user can experiment with the sound and instruments, write mode (Studio/Record)—where, as the user plays an instrument with the keyboard, the system automatically writes down the score, and play mode (Overlay)—where the user can have the system play the score on any instrument or combine different scores to create a song. Potential risks include overwhelming users with complexity of the system or failing to balance flexibility with simplicity, but if addressed well, our system can transform casual system interaction into meaningful musical expression and collaboration between users sharing scores.

Our team will know that our keyboard-driven music system has done well when all users with various music background knowledge can successfully compose a “song” by combining/overlaying created scores and use the system to play the song.

II. User Personas

Group 1: Under 18

Persona 1

			Name: Yu Narukami
			Age: 16
			Occupation: Student
5 Pains • Does not understand how to read or write music. • Music applications can look or feel too complicated. • May not have the time to getting better with the application and learning how to read or write music because of extracurricular activities and trying to solve murders while investigating the TV world. • Does not like tutorials or instructions that are confusing. • Has a tendency to lose interest quickly.	5 Gains • This application helps with reading and writing music. • The application will have a very friendly UX and UI design. • Can practice learning rhythm, scales, and melodies. • Could make practices more fun for the basketball team and soccer team. • Making music could make him more popular in school with other students and teachers.	5 Jobs • Wants to create some beats and tracks for the basketball team and soccer team he is on. • Learning music could look good on a college application or job application. • School work and playing sports can be exhausting so making music can help me relax. • Helps him learn about technologies out there and what applications can do. • Learning about music could help him get a job as a DJ.	

Persona 2

	Name: Chie Satonaka	
	Age: 16	
	Occupation: Student	
5 Pains • Does not have a lot of money to pay for music applications, instruments or music lessons. • No access to music making equipment. • Not very good with technology so could be challenging to get music files to friends and family. • Does not have a lot of knowledge in music history, the different kinds of music, or what is currently happening in music popular culture. • The learning curve for this application may be difficult.	5 Gains • The application has the ability to make sounds that replicate different physical musical instruments very closely. • Can change pitch and duration of note with shift key and spacebar. • Could help me and my friends start a band. • The music I create will make me a better dancer. • Can help advance my technology skills.	5 Jobs • Wants to share musical creations with friends. • Making music gives me something to talk about with my family and friends so this is good for bettering my communication skills. • Will help me learn about the many different kinds or genres of music. • Hopes to make music that friends and self can dance to. • Will help me advance my creativity and imagination when making music.

Persona 3

		Name: Ren Amamiya
		Age: 17
		Occupation: Music Student
5 Pains • The application does not have a touchscreen. • Practicing with musical instruments is loud and disturbs family members at home and the neighbors. • Not enough money to buy a violin or piano. • Cannot practice using the application with other students like practicing with instruments with other students during music class. • Not enough control over audio filters for bass, treble, brilliance, presence, distortion, amplifier, etc. while using musical applications.	5 Gains • The application allows writing musical melodies and scores. • Can practice music and what it might sound like while traveling like a bus or plane. • Using the application, I can practice music at home while not disturbing neighbors using the application with headphones while real musical instruments disturb those at home and the neighbors. • This application will help me become a musical composer. • I can save and edit the music I create.	5 Jobs • With this application, I can build a music portfolio. • I can share my musical compositions online with other composers. • Can combine musical instrument sounds together to make a symphony or orchestra with the application. • With the knowledge and experience I have in music and using this application I have a better chance to have a career making music in movies. • Could help me get a job as a video game composer.

Group 2: 18-40

Persona 4

	Name: Jared	
	Age: 20	
	Occupation: University Student studying Music	
5 Pains • Professional music software is expensive • Hard to collab on music without being in the same room • Has a hard time writing sheet music • Limited budget due to being a full-time student • Switching between practice, writing and playing music is tedious	5 Gains • Born in the digital age • Learns new software quickly • Feels accomplished when creating new music • Plays multiple instruments • Can share creations with peers for feedback	5 Jobs • Experiment with instruments he doesn't own • Turn melodies in his head into scores quickly • Combine parts to prototype band songs • Complete coursework • Play music with friends

Persona 5

			Name: Rebecca		
			Age: 31		
			Occupation: HR Specialist		
5 Pains • Has no musical training • Does not know where to start • Limited free time • Feels embarrassed about being a beginner • Does not know music theory		5 Gains • Organized and detail oriented in relation to her job • Enjoys new hobbies • Comfortable with web apps • Finds music relaxing • Like sharing creative projects on social media		5 Jobs • Find creative outlet outside of work • Learn basic music concepts without a class • Share something creative/fun on social media • Unwind after work • Create a simple melody she would be proud of	

Persona 6

			Name: Christopher		
			Age: 35		
			Occupation: Film Composer		
5 Pains • Some music software is overkill • Hard to share music ideas with directors who don't read notation • Needs to test how different instrument combinations sound like quickly • On set or can't carry full studio setup • Existing tools don't let him quickly swap instruments on a draft		5 Gains • Knowledge in music theory • Familiar with music software • Can quickly sketch music ideas • Can prototype on a laptop anywhere • Values collaborating with others on this team (directors, musicians) remotely		5 Jobs • Rapidly prototype a film cue between meetings • Layer instrument arrangements to demo to a director • Experiment with unfamiliar instruments • Share drafts with collaborators for feedback • Capture musical ideas before they are forgotten	

Group 3: 41-65

Persona 7

	Name: Jemma Burton	
	Age: 53	
	Occupation: High school band director	
5 Pains • Students have varying skill levels, making it hard to teach composition equally • Traditional notation software is complex and time-consuming to teach • Limited class time to demonstrate multiple instruments • Difficulty getting students who are not interested in classical methods to engage • Difficulty coordinating ensemble practice when students cannot all rehearse together	5 Gains • Simple interface allows students to compose without prior theory knowledge • Can demonstrate multiple instruments instantly in class • Encourages student creativity and experimentation • Easy sharing of student compositions for feedback or grading • Ability to simulate a full band without needing all instruments present	5 Jobs • Teach students the basics of music composition. • Provide an engaging, interactive classroom activity • Demonstrate how different instruments sound together • Review and assess student-created music • Facilitate group collaboration on musical pieces

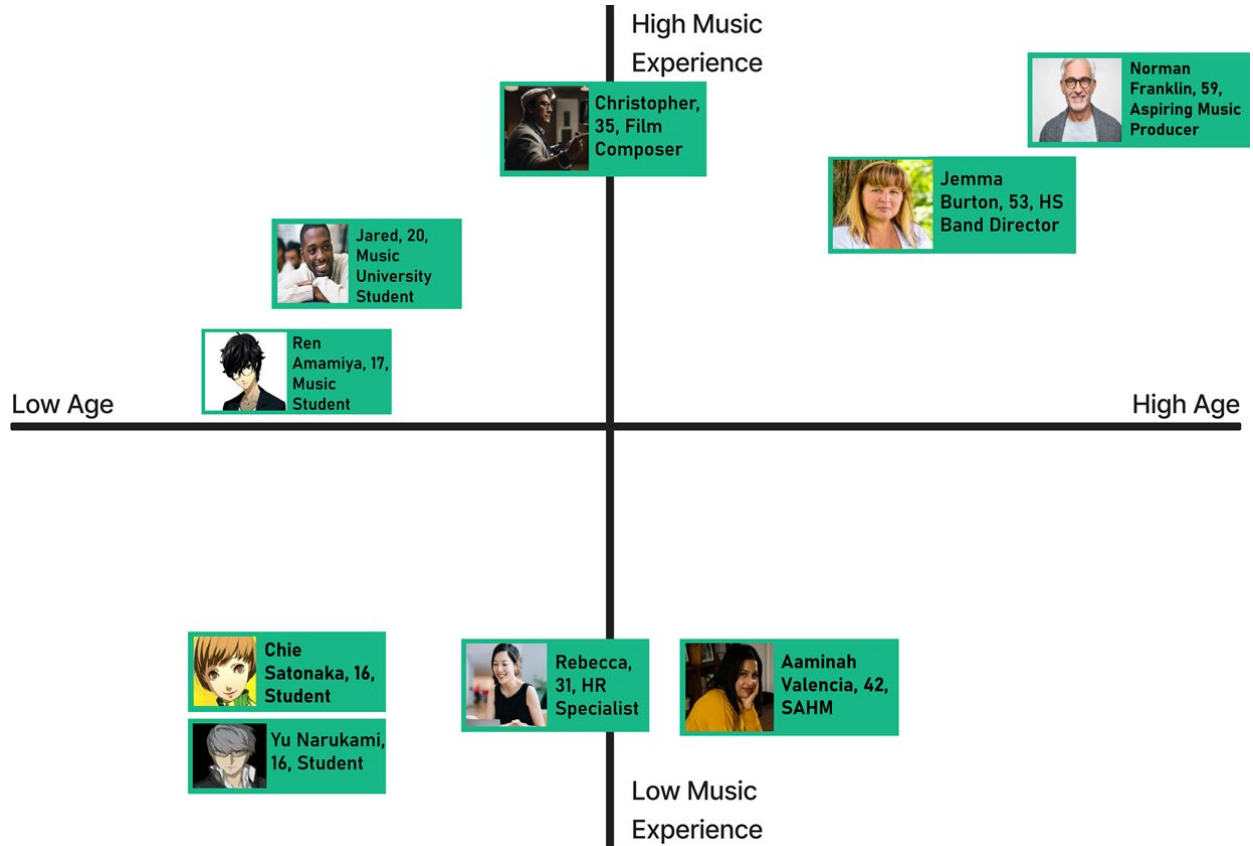
Persona 8

	Name: Aaminah Valencia	
	Age: 42	
	Occupation: Stay-at-home mother	
5 Pains • Has limited time to learn traditional instruments or software • Has a lack of formal music training or knowledge • Difficulty finding engaging creative activities for herself or her children • Frustration with complicated music apps or tools • Limited access to physical instruments at home	5 Gains • Easy, low-barrier way to create music casually • Fun activity she can do with her children • No need for prior music knowledge • Ability to quickly create and share simple songs • Access to multiple instruments without purchasing them	5 Jobs • Entertain and engage herself and her children • Explore a creative hobby without a steep learning curve • Create simple songs or melodies • Share creations with family or friends • Use the app as a casual, stress-free activity

Persona 9

		Name: Norman Franklin
		Age: 59
		Occupation: Aspiring music producer
5 Pains • Professional music software can be expensive and complex • Difficulty quickly prototyping musical ideas • Limited access to a wide range of instruments • Workflow interruptions when switching tools or instruments • Steep learning curve for advanced production tools	5 Gains • Rapid idea prototyping using keyboard input • Ability to experiment with multiple instruments (quickly and easily) • Quick layering of tracks to simulate a full production • Portable and accessible tool (the PWA) • Easy sharing of drafts or concepts with collaborators	5 Jobs • Prototype and sketch musical ideas quickly • Experiment with different instrument combinations • Layer tracks to build more complex compositions • Share drafts with collaborators or peers • Practice and refine music production skills

User Persona Map



III. User Needs

Ranked User Pains

The top-ranking user pains tended to be related to expense and difficulty reading/writing professional music. In particular:

- Professional music software (app or equipment) can be expensive and complex
- Lack of musical training or background knowledge
- Difficulty reading or writing sheet music
- Difficulty collaborating with other musicians (or testing different instruments)
- Limited time to practice

Ranked User Gains

The top-ranking user gains tended to be related to easy access to different instruments/sounds. In particular:

- Easy access to different instruments and sounds without cost
- Easy access to music creation and collaboration
- Casual way to learn music concepts
- Simple UI

Ranked User Jobs

The top-ranking user needs tended to be related to ability to share music/collaborate with others on the app. In particular:

- Ability to share music/collaborate with others on the app
- Quickly create basic music or prototype songs by combining created sounds
- Experiment with different sounds

IV. Product Features

Ranked Gain Creators

The top-ranking features that the PWA must have that create user gains are:

- Keyboard-based instrument interaction with multiple sound options – **REQUIRED**
 - This enables users to easily access and experiment with different instruments and sounds without cost.
- Three-mode system (Test, Write, Play) – **REQUIRED**
 - This provides a structured yet simple workflow for experimentation, automatic score creation, playback, and sharing.
- Automatic score generation while playing (Write Mode) – **REQUIRED**
 - This allows users to create music without needing prior knowledge of sheet music, lowering the barrier to entry.
- Score playback on different instruments and layering (Play Mode) – **REQUIRED**
 - This supports combining multiple scores to simulate a "band" or "song," enhancing creativity and experimentation.
- Simple and intuitive user interface – **REQUIRED**
 - This ensures accessibility for users with little to no musical background.
- Basic sharing functionality for created scores – **REQUIRED**
 - This allows users to share their music with others, supporting collaboration and user engagement.

The top-ranking optional features that would create user gains are:

- Real-time collaboration on shared scores – **OPTIONAL**
 - This would enhance collaborative creation but is not essential for initial deployment.
- Guided learning prompts or hints for music concepts – **OPTIONAL**
 - This would provide a casual way to learn music theory while using the system.

Ranked Pain Relievers

The top-ranking features that the PWA must have that relieve user pains are:

- Free access via web-based platform (no expensive software/equipment required) – **REQUIRED**
 - This eliminates the cost barrier associated with professional music tools.
- No prior music knowledge required (automatic translation of input to scores) – **REQUIRED**
 - This reduces difficulty in reading and writing sheet music.
- Simplified music creation process through keyboard input – **REQUIRED**
 - This removes complexity found in traditional music software.
- Predefined instrument mappings and sound presets – **REQUIRED**
 - This prevents users from being overwhelmed and simplifies interaction with different instruments.

- Quick music prototyping through score layering and playback – **REQUIRED**
 - This addresses limited time to practice by enabling rapid creation.
- Asynchronous sharing (send and play scores without needing simultaneous users) – **REQUIRED**
 - This reduces collaboration barriers when users are not available at the same time.

The top-ranking optional features that would relieve user pains are:

- Advanced customization of instruments and sound parameters – **OPTIONAL**
 - This would be useful for experienced users but not necessary for beginners.
- Live collaboration with multiple users editing simultaneously – **OPTIONAL**
 - This would address collaboration challenges but adds system complexity.

Ranked Products and Services

The top-ranking features that the PWA must have to do jobs are:

- Play instruments using a keyboard interface – **REQUIRED**
 - Core functionality that enables users to interact with the system.
- Experiment with sounds in Test Mode – **REQUIRED**
 - This allows users to explore instruments freely without pressure.
- Automatically create musical scores in Write Mode – **REQUIRED**
 - This supports quick and intuitive music creation.
- Play back and modify scores in Play Mode – **REQUIRED**
 - This enables users to refine and experience their compositions.
- Combine multiple scores to create layered compositions ("band"/"song" feature) – **REQUIRED**
 - This fulfills the need to prototype songs and simulate collaboration.
- Save and share created scores with others – **REQUIRED**
 - This supports the primary job of collaboration and distribution.

The top-ranking optional features that would do jobs are:

- Real-time collaborative editing environment – **OPTIONAL**
 - This would enhance teamwork but can be deferred to later versions.

V. Acceptance Tests

Test 1A

Test Card Strategyzer

Learning to Read and Write Music Test Before release of PWA

Users with no ability to read or write music. 1 day

STEP 1: HYPOTHESIS

We believe that users will be able to navigate the prototype to hypothetically create music

Critical:

STEP 2: TEST

To verify that, we will ask users question where they are required to interact and navigate through the low-fidelity prototype

Test Cost: Data Reliability:

STEP 3: METRIC

And measure the accuracy of how the user interacts with the prototype when prompted in comparison to how the user was supposed to interact

Time Required:

STEP 4: CRITERIA

We are right if the user gets at least 80% of the interactions correct

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Test Completion Steps:

1. Let users with minimal to no ability to read or write music use our application (low-fidelity prototype).
2. Have users spend 1 day using our application.
3. Give users a test to see if they can correctly interact with the prototype.
4. Test is successful if users get at least 80% of the questions correct.

Testing:

Laurelle chose to use the design method [5.11 Behavioral Prototype](#) when conducting her test. She identify the situation to be simulated: The situation she chose to better understand user behaviors is a trial of the application via our low fidelity prototype. She prepared the environment for simulation and presented the user with the 4 pages in a way that is easy to read and understand. Make accommodations (print out) if a screen will be difficult. She allowed the user to view each page one by one before asking any questions and let them make their own observations to prepare for the test.

Questions:

1. How would you start making music?
2. Where would you go if you want to test an instrument?
 - a. If you wanted to hear a Trumpet sound what button would you click?
3. Where would you go if you want to change your instrument?
4. What button can you click on the '(Instrument Being Played) Studio' page that will allow you to start recording the sounds you make?
5. Where would you go if you want to overlay another score?
6. Where would you go if you want to play your score?
7. That is all the questions I have for you. Are there any questions you have for me or things you feel should be changed on this website application?

Answers:

1. I would click on the start making music button.
2. Depending on the instrument, I would click the corresponding button.
 - a. I would click the trumpet button.
3. I would click the change instrument button.
4. I would click the record button.
5. The user had a little difficulty finding the overlay button on the recording screen.
6. I would click first track, and I would like an option to choose files or record a track. Then click second track. Then I would click the begin overlay process button.
7. Too many colors, no consistency in design, doesn't show you which page you are currently on.

Conclusion: I would consider this test successful as the user was able to find most buttons and options to navigate the page with ease. She had some trouble with the navbar and she mentioned that it would be helpful if it showed which page you were currently on. She wasn't sure which page was the homepage because of the navbar. She would have also appreciated more consistency in the design of the pages.

Test 1B

Test Card		Strategyzer	
Learning to Read and Write Music Test		Before release of PWA	
Users with no ability to read or write music.		1 day	
STEP 1: HYPOTHESIS			
We believe that		users will be able to navigate the prototype to hypothetically create music	
		Critical:   	
STEP 2: TEST			
To verify that, we will		ask users question where they are required to interact and navigate through the low-fidelity prototype	
		Test Cost:    Data Reliability:   	
STEP 3: METRIC			
And measure		the accuracy of how the user interacts with the prototype when prompted in comparison to how the user was supposed to interact	
		Time Required:   	
STEP 4: CRITERIA			
We are right if		the user gets at least 80% of the interactions correct	

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Test Completion Steps:

1. Let users with minimal to no ability to read or write music use our application (low-fidelity prototype).
2. Have users spend 1 day using our application.
3. Give users a test to see if they can correctly interact with the prototype.
4. Test is successful if users get at least 80% of the questions correct.

Testing: For the low-fidelity prototype test, I (Greg) went with a Usability Test to see if the website application was set up in a way that was easy to navigate and not difficult to understand how to create music, record music, and overlay tracks. I did the test with a participant who was a male, 70 years old with low technology skills and no ability to read or write music in an environment they felt comfortable doing the test. I gave the participant four pieces of paper that are pages for our low-fidelity prototype of the music website application we are building and had them read what was on each of the pages. I then asked the participant to study the pages for a few minutes, so the participant knew what this website was going to do and how to navigate the website.

After the participant spent a few minutes looking at the low-fidelity prototype pages I asked if the participant if they were ready to answer several open ended questions I had about this music website application my team was creating, and they said they were ready for the questions.

For this test to be a success, the participant must understand how to do 90% of the questions I give about the website application. Below are the questions I asked the participant along with their responses:

Questions:

1. "Which one of these pages is the Home page? Please point to the Home page."
2. "What button would you press on the Home page that would get you to the next page to start making music?"
3. "When you click the Start Making Music button what page do you think it would take you to?"
4. "When you are on the Select Your Sound page what sounds can make with this website application?"
5. "If you wanted to hear a Trumpet sound what button would you click?"
6. "After you click one of those button what page would you go to next?"
7. "When on the (Instrument Being Played) Studio page what will '(Instrument Being Played) Studio' say if on the previous page you clicked Piano?"
8. "When on the (Instrument Being Played) Studio page what will 'Use your keyboard to make (Instrument) sounds' say if on the previous page you clicked Violin?"
9. "What button can you click on the '(Instrument Being Played) Studio' page that will allow you to start recording the sounds you make?"
10. "On the '(Instrument Being Played) Studio' page where it says 'Musical Score Sheet (Read Only)' what do you think happens when you start pressing keys on your keyboard to make sounds?"
11. "If you were making a recording how would you stop your recording or what button would you click?"
12. "When on the '(Instrument Being Played) Studio' page what would you click if you wanted to change your instrument?"
13. "If you created one music track with one instrument and another music track with another instrument where would you go to overlay those two tracks?"
14. "When on the 'Overlay Tracks' page what would you click to put in your first track?"

15. "When on the 'Overlay Tracks' page what would you click to put in your second track?"
16. "After both of your tracks have been put into both the 'First track to overlay' and 'Second track to overlay' positions what button do you click to begin the overlay process?"
17. "That is all the questions I have for you. Are there any questions you have for me or things you feel should be changed on this website application?"

Answers:

1. Participant points to the correct page that was the Home page. User said, "It is the Home page because it had the necessary information somebody would see when you first visit a website."
2. Participant points to the Start Making Music button on the Home page and says, "This is the button you would press because it says start making music."
3. Participant points to the Instrument page or Select Your Sound page.
4. Participant says, "Acoustic Guitar, Piano, Violin, Trumpet."
5. Participant points to Trumpet button. (The user also knew what button to press for hearing the other sounds as well.)
6. Participant points to the (Instrument Being Played) Studio page.
7. Participant says, "It would say Piano Studio."
8. Participant says, "It would say 'Use your keyboard to make Violin sounds'."
9. Participant says, "I would click the Record button or it looks like I can press the Enter button on my keyboard."
10. Participant says, "I think that the image of a musical score sheet appears there with the lines and notes on the lines of the keys I am pressing."
11. Participant says, "Well obviously the Stop Recording button."
12. Participant says, "I would click where it says 'Change Instrument'."
13. Participant points to the 'Overlay Tracks' page.
14. Participant says, "I would click the 'First track to overlay' button."
15. Participant says, "I would click the 'Second track to overlay' button."
16. Participant says, "I would click the 'Begin Overlay Process' button."
17. Participant asked interviewer, "What happens when I get done with the recording? Where does the recording go? Can I download the musical score sheet after the recording? I would like to hear what it sounds like when it is finished."

Results: The results from the Usability Test are that the participant knew how to navigate the website application. The participant understood how to use the website application to select the instrument they wanted to hear sound from, how to record music, how to change instruments, and how to overlay tracks they have recorded. The participant did want to know what would happen when they finished recording their music. They wanted to know if the file would automatically go to the Downloads directory or some other folder or directory on their computer. The participant wanted to know if maybe a link would show up on the website application where they had the option of downloading the music recording file they just created. The participant wanted to know the same thing about the musical score sheet once the recording has been stopped and if the musical score sheet would go automatically to the Downloads directory or if a link would come up to download the musical score sheet.

Conclusion: I think the music creating website application is currently well designed for users to navigate and create music. The reason is because the participant is not very tech savvy and is not musically literate but correctly understood how to do 100% of the questions I gave them about the low-fidelity website application.

The problems I see currently with developing this website application are what happens when users get done with recording their music. Will the users' music files go directly to the Downloads directory or become a link once the users stop their recordings. Also, what will users be able to download the musical score sheets once they stop their recording or will the musical score sheets be automatically put in the Downloads directory. There are changes that need to be made to the design of this website application to give users the information needed about types of music files created by this application and what types of files the application is capable of overlaying.

Test 1C

Test Card		Strategyzer	
Learning to Read and Write Music Test		Before release of PWA	
Users with no ability to read or write music.		1 day	
STEP 1: HYPOTHESIS			
We believe that	users will be able to navigate the prototype to hypothetically create music		
		Critical:	
STEP 2: TEST			
To verify that, we will	ask users question where they are required to interact and navigate through the low-fidelity prototype		
	Test Cost:	Data Reliability:	
			
STEP 3: METRIC			
And measure	the accuracy of how the user interacts with the prototype when prompted in comparison to how the user was supposed to interact		
		Time Required:	
STEP 4: CRITERIA			
We are right if	the user gets at least 80% of the interactions correct		
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Test Completion Steps:

1. Let users with minimal-to-no ability to read or write music use our application (low-fidelity prototype).
2. Have users spend 1 day using our application.
3. Give users a test to see if they can correctly interact with the prototype.
4. Test is successful if users get at least 80% of the questions correct.

Testing: This was another Usability Test I (Greg) did to see if the website application was set up in a way that was easy to navigate and not difficult to understand how to create music, record music and overlay tracks. This test was done on a male who was 12 years old with moderate technology skills and no ability to read or write music in an environment they felt comfortable doing the test.

I gave the participant four pieces of paper that are pages for our low-fidelity prototype of the music website application we are building and had them read what was on each of the pages. I then asked the participant to study the pages for a few minutes, so the participant knew what this website was going to do and how to navigate the website. After the participant spent a few minutes looking at the low-fidelity prototype pages I asked if the participant if they were ready to answer several open-ended questions I had about this music website application my team was creating, and they said they were ready for the questions.

For this test to be a success the participant must understand how to do 90% of the questions I give about the website application. Below are the questions I asked the participant along with their responses:

Questions:

1. "Which one of these pages is the Home page? Please point to the Home page."
2. "What button would you press on the Home page that would get you to the next page to start making music?"
3. "When you click the Start Making Music button what page do you think it would take you to?"
4. "When you are on the Select Your Sound page what sounds can make with this website application?"
5. "If you wanted to hear a Trumpet sound what button would you click?"
6. "After you click one of those button what page would you go to next?"
7. "When on the (Instrument Being Played) Studio page what will '(Instrument Being Played) Studio' say if on the previous page you clicked Piano?"
8. "When on the (Instrument Being Played) Studio page what will 'Use your keyboard to make (Instrument) sounds' say if on the previous page you clicked Violin?"
9. "What button can you click on the '(Instrument Being Played) Studio' page that will allow you to start recording the sounds you make?"
10. "On the '(Instrument Being Played) Studio' page where it says 'Musical Score Sheet (Read Only)' what do you think happens when you start pressing keys on your keyboard to make sounds?"
11. "If you were making a recording how would you stop your recording or what button would you click?"
12. "When on the '(Instrument Being Played) Studio' page what would you click if you wanted to change your instrument?"
13. "If you created one music track with one instrument and another music track with another instrument where would you go to overlay those two tracks?"
14. "When on the 'Overlay Tracks' page what would you click to put in your first track?"
15. "When on the 'Overlay Tracks' page what would you click to put in your second track?"
16. "After both of your tracks have been put into both the 'First track to overlay' and 'Second track to overlay' positions what button do you click to begin the overlay process?"
17. That is all the questions I have for you. Are there any questions you have for me or things you feel should be changed on this website application?

Answers:

1. Participant points to the correct page and says, "It is this one."
2. Participant points to the Start Making Music button and says, "You would click this button."
3. Participant points to the Select Your Sound page and says, "This one."
4. Participant says, "Acoustic Guitar, Piano, Violin, Trumpet."
5. Participant points to Trumpet button. (The user also knew what button to press for hearing the other sounds as well.)
6. Participant points to the (Instrument Being Played) Studio page.
7. Participant says, "It would say Piano Studio."
8. Participant says, "It would say 'Use your keyboard to make Violin sounds'."
9. Participant says, "I would click the Record button or it looks like I can press the Enter button on my keyboard."
10. Participant says, "It would turn into a musical score sheet where notes would come up with the keys I press on the keyboard."
11. Participant says, "The Stop Recording button."
12. Participant points to the "Change Instrument" button.
13. Participant points to the "Overlay Tracks" page and says, "This one."
14. Participant points to the "First track to overlay" button and said, "I would click here."
15. Participant points to the "Second track to overlay" button and said, "I would click here."
16. Participant points to the "Begin Overlay Process" button and says, "This button."
17. The participant had questions about what kind of files can you put in the overlay process. The participant also wanted to know what kind of music files the application will make and will they be able to save the musical score sheet. Participant says this website looks good and would like to try it out when it is completed.

Results: The results from this Usability Test was that the participant knew how to navigate the website application. The participant understood how to use the website application to select the instrument they wanted to hear sound from, how to record music, how to change instruments, and how to overlay tracks they have recorded. The participant did have questions about how files are created using this website application and what types of files can be used to overlay tracks.

Conclusion: I think from this test the website is shown to have an intuitive design and can be easily navigated by younger people. The reason is because this participant correctly understood how to do 100% of the questions I gave them about the low-fidelity website application. The younger participant had questions about how does the music someone creates get saved, what types of music files can you put in the overlay process, and how to save the musical score sheets once the recording stops. There are changes that need to be made to the design of this website application to give users the information needed about types of music files created by this application and what types of files the application is capable of overlaying.

Test 1D

Test Card		Strategyzer	
Learning to Read and Write Music Test		Before release of PWA	
Users with no ability to read or write music.		1 day	
STEP 1: HYPOTHESIS			
We believe that	users will be able to navigate the prototype to hypothetically create music		
			Critical: 
STEP 2: TEST			
To verify that, we will	ask users question where they are required to interact and navigate through the low-fidelity prototype		
		Test Cost: 	Data Reliability: 
STEP 3: METRIC			
And measure	the accuracy of how the user interacts with the prototype when prompted in comparison to how the user was supposed to interact		
			Time Required: 
STEP 4: CRITERIA			
We are right if	the user gets at least 80% of the interactions correct		
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Test Completion Steps:

1. Let users with minimal-to-no ability to read or write music use our application (low-fidelity prototype).
2. Have users spend 1 day using our application.
3. Give users a test to see if they can correctly interact with the prototype.
4. Test is successful if users get at least 80% of the questions correct.

Testing:

The design method use was “Design Method Cards: Usability Testing.” The participant (Mia, 23yo female, someone with little or no understanding of music production, software or music knowledge) was given context, and the tester (Allan) explained what the application is about.

The tester then let them try to view the prototype to prime them for the upcoming steps. The participant was then to work through a user workflow:

1. Home Page
2. Instrument Selection Page
3. Studio Page
4. Overlay Tracks Page

Then we then asked multiple open-ended questions, with the responses to them seen below. Overall, Mia was given the low-fidelity prototype pages and was asked to study them for 5 minutes.

Questions:

1. Which page is the Home Page? Can you point to it?
2. What button on the Home Page would you press to start making music?
3. After clicking that button, where do you go and what sounds can you make?
4. Let's say you picked Piano and you're in the studio. How do you start recording?
5. You have two recorded tracks. Where do you go to combine them and what do you click?
6. Scenarios and Tasks
 - a. Identifying the multitude of pages
 - b. Selection of an instrument
 - c. Start recording in the Studio
 - d. Navigation to the overlay page and combine two recorded tracks

Answers:

1. The user correctly identified the Home page.
2. The user pointed to the 'Start Making Music' button.
3. The user correctly identified the 'Select your Sound button'.
4. The user selected the record button.
5. The user clicked on the Overlay page and put one track in each slot although was confused where the tracks come from and if they could 'download' them.

Conclusion: 5/5 questions were correct, so the testing was successful. The participant commented that the workflow between the studio and overlay tracks was ambiguous but overall, the application was clear. The flow between Studio and Overlay tracks page needs to be less ambiguous. The ambiguity comes from a lack of understanding of where track storage and download.

Test 2A

Test Card		Strategyzer	
Ease of Use Test	Before release of PWA		
Whole group	1 week		
STEP 1: HYPOTHESIS			
We believe that		when users create music with our application and have to overlay a musical instrument track on top of another track the UI design and UX design will not confuse users how to do that and if fact will be easy to do for most user.	
		Critical: 	
STEP 2: TEST			
To verify that, we will		have users spend time using the application and have users create a musical instrument track and then have users create another musical instrument track and see if users can figure out how to overlay one track on top of the other.	
		Test Cost:  Data Reliability: 	
STEP 3: METRIC			
And measure		the percentage of users that can figure to overlay one track on top of another track.	
		Time Required: 	
STEP 4: CRITERIA			
We are right if		at least 80% of users can figure to overlay a track all by themselves.	
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Test Completion Steps:

1. Have users make music with our application.
2. Have users spend 1 week using our application and creating at least two tracks of music with at least two different instruments.
3. Give users a test to see if they can find out how to overlay one track on top of another track.
4. The test is successful if 80% of users can figure out how to do this themselves.

Design Method: Usability Testing

Testing: Participant was Mia, 22yr old college student with little to no music knowledge. The task given to the participant: Given one hour, the participant was told to create two overlaid tracks using two different instruments. The time to achieve overall task was ~40 minutes. Mia

had tried to record both instruments simultaneously assuming that would layer them. Noticed the navbar overlay page via the navbar.

Questions:

1. Was there anything that helped you figure out how to overlay one track on top of another?
2. Was there anything confusing when using the application when trying to make your music?
3. Have you spent time using other kinds of music creating applications that may be similar to our application?
4. Did you think the UI design and UX design of the application was good?
5. Are there any improvements for the application you would like to see?

Answers:

1. The participant described that the overlay page was self-describing.
2. The participant mentioned confusion in the keyboard layout. They weren't sure what buttons to press since they didn't know understand the note names. The user mentioned they were initially confused with the flow between Studio and Overlay since they had to manually download and export the song to overlay.
3. The participant hasn't spent any time in other music creation applications.
4. The participant mentions that the UI/UX design of the application is good and noted they liked the dark theme it encapsulated. The transition between Studio and Overlay was the only hiccup.
5. A tutorial in the workflow could make the application clearer.

Conclusion: From my (Allan) understanding, many criticisms can be fixed by using a tutorial, which we did end up making before I conducted this interview. Keyboard layout could be fixed maybe by a small introduction over notes and color-coding keys.

Test 2B

Test Card		Strategyzer	
Ease of Use Test	Before release of PWA		
Whole group	1 week		
STEP 1: HYPOTHESIS			
We believe that		when users create music with our application and have to overlay a musical instrument track on top of another track the UI design and UX design will not confuse users how to do that and if fact will be easy to do for most user.	
		Critical: 	
STEP 2: TEST			
To verify that, we will		have users spend time using the application and have users create a musical instrument track and then have users create another musical instrument track and see if users can figure out how to overlay one track on top of the other.	
		Test Cost:  Data Reliability: 	
STEP 3: METRIC			
And measure		the percentage of users that can figure to overlay one track on top of another track.	
		Time Required: 	
STEP 4: CRITERIA			
We are right if		at least 80% of users can figure to overlay a track all by themselves.	
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Test Completion Steps:

1. Have users make music with our application.
2. Have users spend 1 week using our application and creating at least two tracks of music with at least two different instruments.
3. Give users a test to see if they can find out how to overlay one track on top of another track.
4. The test is successful if 80% of users can figure out how to do this themselves.

Design Method: DI Method Cards: Feedback Capture Matrix. This is a structured way of gathering and organising feedback from users. The feedback is organized into 4 groups: “What worked”, “What can be improved”, “Questions”, and “Ideas”. Feedback is clustered and put in a 2x2 matrix in the *Conclusion* section.

Testing: Participant was a 48 yr female with no previous music experience.

Testing: Participant was Mia, 22yr old college student with little to no music knowledge. The task given to the participant: Given one hour, the participant was told to create two overlayed tracks using two different instruments. The time to achieve overall task was ~40 minutes. Mia had tried to record both instruments simultaneously assuming that would layer them. Noticed the navbar overlay page via the navbar.

Questions:

1. Was there anything that helped you figure out how to overlay one track on top of another?
2. Was there anything confusing when using the application when trying to make your music?
3. Have you spent time using other kinds of music creating applications that may be similar to our application?
4. Did you think the UI design and UX design of the application was good?
5. Are there any improvements for the application you would like to see?

Answers:

1. Overlay page mechanics were straightforward and easy.
2. User did not expect to use the keyboard to type and had trouble seeing the directions to use the keyboard instead of clicking.
3. This user has zero experience with other music applications.
4. Partly. The colors looked nice and there were no distractions. The user did have some issues finding directions and the tutorial.
5. The user would prefer to be able to use the sliders to trim down the actual music sample and listen to the file while trimming it down.

Conclusion:

Feedback Matrix	
What Worked • Tutorial was very helpful • UI design was simple and intuitive	What Can Be Improved • Trimming could be improved via the sliders
Questions • No questions	Ideas • Would like to listen to individual samples while trimming • Message pop-up if a user tries to click keys instead of typing

Test 3

The Test Card		Strategyzer
Test name: Music Creating Difficulty Test	Duration: Before release of PWA	
Assigned to: Whole group	Deadline: up to 3-5 days	
STEP 1: HYPOTHESIS		
We believe that users will have an easy and enjoyable time creating music with out application.		
STEP 2: TEST		
To verify that, we will have users spend time creating music with this application.		
STEP 3: METRIC		
And measure	the % of users that say creating music with our application was easy to learn and fun to use.	
STEP 4: CRITERIA		
We are right if at least 75% of users say that the application was easy to learn and had fun making music on the application.		
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Test Completion Steps:

1. Have users with some musical experience make music with our application.
2. Have users spend up to 3-5 days using our application and creating music songs and scores.
3. Give users a product test or consumer test asking question like:
 - a. “Do you think positively or negatively about this application?”
 - b. “How easy was it to use this product?”
 - i. 1. Very Difficult
 - ii. 2. Somewhat Difficult
 - iii. 3. In the middle
 - iv. 4. Somewhat Easy
 - v. 5. Very Easy
 - c. “How fun was it to use this product?”

- i. 1. Complete opposite of fun
 - ii. 2. Not really that fun
 - iii. 3. In the middle
 - iv. 4. Somewhat Fun
 - v. 5. Very Fun
- 4. Test is successful if 80% of users think positively of our product and there are scores of 4 or above for the questions “How easy to use” and “How fun”.

Design Method:

DI Method Cards: Feedback Capture Matrix. This is a structured way of gathering and organizing feedback from users. The feedback is organized into 4 groups: “What worked”, “What can be improved”, “Questions”, and “Ideas”. Feedback is clustered and put in a 2x2 matrix in the *Conclusion* section.

Testing: Testing was proctored by Laurelle. Participant was given the prototype on a computer and told to make a song using any instruments and to make sure to overlay multiple tracks. No further instructions about the site’s functionality were given.

Questions:

1. “How easy was it to use this product?”
 - a. 1. Very Difficult
 - b. 2. Somewhat Difficult
 - c. 3. In the middle
 - d. 4. Somewhat Easy
 - e. 5. Very Easy
2. “How fun was it to use this product?”
 - a. 1. Complete opposite of fun
 - b. 2. Not really that fun
 - c. 3. In the middle
 - d. 4. Somewhat Fun
 - e. 5. Very Fun
3. Why did you find this application easy to use?
4. What was fun about using this application?
5. Was there anything confusing when using the application when trying to make your music?
6. Have you spent time using other kinds of music creating applications that may be similar to our application?
7. Did you think the UI design and UX design of the application was good?
8. Are there any improvements to the application you would like to see?

Answers:

1. Somewhat easy (4)
2. Somewhat fun (4)
3. The layout was very simple, didn’t have any issues knowing where to go.
4. Making the music was the fun part.

5. Somewhat confused on how to trim music and playing and stopping individual tracks.
6. No previous experience.
7. The pages looked pretty good.
8. Would prefer to be able to type in metronome instead of having to slide to a specific number.
The stop button on the overlay page doesn't stand out. No indication on individual tracks which track is playing. No way to know how far along the recording is/ no way to go back.

Conclusion:

Feedback Matrix	
What Worked - Overall layout of the website - Edit option on overlay page showing sound waves - Metronome was helpful and simple to use	What Can Be Improved - How users trim a track (didn't read the caption, immediately started dragging to trim) - Stop option for individual tracks - Metronome
Questions No questions	Ideas - Allow metronome number to be typed in - Change play button on individual tracks to stop once playing - Change color of stop button on bottom to be more noticeable - Add sliding bar to show progress of tracks playing

Test 4A

Test Card



Quality of Sound Plus Other Benefits Test	Before PWA release
User with moderate to high music creations ability	Up to 3-5 days

STEP 1: HYPOTHESIS

We believe that users will think the instrument they choose to hear sound from is very similar to the sound created from the real physical instrument. Users will also like the benefits of being able to make music that sounds like real physical instruments without disturbing others and collaborating with other musicians.

Critical: 

STEP 2: TEST

To verify that, we will have real musicians (or those with minimum moderate music creation abilities) and those in the music industry try out our application.

Test Cost:  Data Reliability: 

STEP 3: METRIC

And measure the percentage of users that say the sound is very similar to that of real physical instruments and the benefits of making music with the application is necessary in the music making world.

Time Required: 

STEP 4: CRITERIA

We are right if at least 75% of users say the application has sound similar to real physical instruments and the benefits of this application are worth using this music making application.

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Test Completion Steps:

1. Have users make music with our application.
2. Have users that have musical experience spend up to 3-5 days using our application and creating music with our application.
3. Give users questions like:
 - a. “How close is the sound to actual physical musical instruments?”
 - i. 1. Not close at all
 - ii. 2. Somewhat not close
 - iii. 3. In the middle
 - iv. 4. Pretty close

- v. 5. Very close
 - b. “How important is it to you knowing you can create music without disturbing your friends, family or neighbors?”
 - i. 1. Not important at all
 - ii. 2. Somewhat not important
 - iii. 3. In the middle
 - iv. 4. Kind of important
 - v. 5. Extremely Important
 - c. “How important is it to you knowing you can collaborate with other musicians when using this application and share the music you create with other musicians online?”
 - i. 1. Not important at all
 - ii. 2. Somewhat not important
 - iii. 3. In the middle
 - iv. 4. Kind of important
 - v. 5. Extremely Important
- 4. Test is successful if the score of all these questions is a 4 or above from those with musical experience.

Design Method: DI Method Cards: Feedback Capture Matrix. This is a structured way of gathering and organising feedback from users. The feedback is organized into 4 groups: “What worked”, “What can be improved”, “Questions”, and “Ideas”. Feedback is clustered and put in a 2x2 matrix in the *Conclusion* section.

Testing: A short introduction was given online over WhatsApp with a 32yo male participant who describes himself as having "proficient music knowledge." When asked to expand on this, he stated that he can play the piano and can (usually) read sheet music. The participant was then given no more than 16 hours to use the music application. He was told he was required to make a 4 seconds minimum song that combines at least 2 instruments. 16 hours later, a brief interview was given, and the questions seen below were asked. The participant was also given 3 survey questions after the interview to answer, also seen below.

Questions:

1. Did you find this application easy to use or fun to use?
2. What did you think of the sound quality of the application?
3. Was there anything confusing when using the application when trying to make your music?
4. Have you spent time using other kinds of music creating applications that may be similar to our application?
5. Did you think the UI design and UX design of the application was good?
6. Are there any improvements for the application you would like to see?
7. “How close is the sound to actual physical musical instruments?”
 - a. 1. Not close at all
 - b. 2. Somewhat not close
 - c. 3. In the middle
 - d. 4. Pretty close
 - e. 5. Very close

8. "How important is it to you knowing you can create music without disturbing your friends, family or neighbors?"
 - a. 1. Not important at all
 - b. 2. Somewhat not important
 - c. 3. In the middle
 - d. 4. Kind of important
 - e. 5. Extremely Important
9. "How important is it to you knowing you can collaborate with other musicians when using this application and share the music you create with other musicians online?"
 - a. 1. Not important at all
 - b. 2. Somewhat not important
 - c. 3. In the middle
 - d. 4. Kind of important
 - e. 5. Extremely Important

Answers:

1. Participant stated that the application was easy to use for the most part, but wouldn't necessarily describe the application as "fun." He also stated that he did not like the original key bindings and was wondering how he could change those.
2. Participant stated that the sound quality was good. He thought that the piano and maybe the guitar instrument were close enough to what the actual instrument sounds like, but the violin and trumpet could be improved.
3. Participant stated for the most part, he understood how to make music. However, he wished that there was a key binding that could be pressed to start recording, otherwise there was a delay between pressing the record button and getting his hands situated on the keyboard to start playing. Also, he stated that he wished the keys below could also be clicked to play sound. Finally, participant stated that he thinks that start time in the "Overlay" section did not work. He wished that overlaying the songs was more interactive and better implemented.
4. Participant stated that he used GarageBand in the past, otherwise he does not use music applications generally to create music.
5. Participant stated he found the web application design to be good. He stated that it was a little painful to read the pink text "Use the keyboard to make (instrument) sounds (Range: C2 - D6)" in "Studio," however, and wished the color was different.
6. Participant stated that he would like the suggestions he gave for the previous questions to be implemented. He stated that when he would go from the "Overlay" page to any other page then back, the songs he overlayed would disappear. So he wants those songs to stay unless the page is refreshed. He had the same advice for the "Studio" page after the user records some notes (he wanted to be able to redownload the song if he returns to the page). Otherwise, he had no additional suggestions for improvements.
7. 3
8. 3
9. 3

Conclusion:

Feedback Matrix	
What Worked • Was able to successfully make an 8 minute “song” • Current app features worked for the participant • App UI design is overall good • Sound quality is overall good	What Can Be Improved • Violin and trumpet sound could be improved • Allow users to change key bindings (give user option to change current settings) • Participant wanted optional key binding to start/stop recording (maybe enter/spacebar was not working from him?) • Overlay needs to be more interactive/visual? • Wants recorded sheet music in “Studio” and songs in “Overlay” to persist
Questions • Participant had no questions	Ideas • Add user login so users can have their own settings? • Add more sounds for violin/trumpet to improve sound? • Improve Overlay section

Overall, while the test was not necessarily a success, the feedback the participant provided was extremely valuable for developing the high-fidelity prototype. The participant overall seemed to like the music app and thought that the current UI design was good. However, he had multiple criticisms for what could be improved, as seen in the Feedback Matrix. Based on his feedback, we think we could add a user login feature so that users can have their own settings (and ability to change the key bindings), as well as the potential ability to have recordings persist between page changes. We will add more sound files for the violin and trumpet to hopefully improve the sound as suggested, and we will also add more interactive elements in the Overlay section to improve users’ ability to create songs.

Test 4B

Test Card



Quality of Sound Plus Other Benefits Test	Before PWA release
User with moderate to high music creations ability	Up to 3-5 days

STEP 1: HYPOTHESIS

We believe that users will think the instrument they choose to hear sound from is very similar to the sound created from the real physical instrument. Users will also like the benefits of being able to make music that sounds like real physical instruments without disturbing others and collaborating with other musicians.

Critical: 

STEP 2: TEST

To verify that, we will have real musicians (or those with minimum moderate music creation abilities) and those in the music industry try out our application.

Test Cost:  Data Reliability: 

STEP 3: METRIC

And measure the percentage of users that say the sound is very similar to that of real physical instruments and the benefits of making music with the application is necessary in the music making world.

Time Required: 

STEP 4: CRITERIA

We are right if at least 75% of users say the application has sound similar to real physical instruments and the benefits of this application are worth using this music making application.

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Test Completion Steps:

1. Have users make music with our application.
2. Have users that have musical experience spend up to 3-5 days using our application and creating music with our application.
3. Give users questions like:
 - a. “How close is the sound to actual physical musical instruments?”
 - i. 1. Not close at all
 - ii. 2. Somewhat not close
 - iii. 3. In the middle
 - iv. 4. Pretty close

- v. 5. Very close
 - b. “How important is it to you knowing you can create music without disturbing your friends, family or neighbors?”
 - i. 1. Not important at all
 - ii. 2. Somewhat not important
 - iii. 3. In the middle
 - iv. 4. Kind of important
 - v. 5. Extremely Important
 - c. “How important is it to you knowing you can collaborate with other musicians when using this application and share the music you create with other musicians online?”
 - i. 1. Not important at all
 - ii. 2. Somewhat not important
 - iii. 3. In the middle
 - iv. 4. Kind of important
 - v. 5. Extremely Important
- 4. Test is successful if the score of all these questions is a 4 or above from those with musical experience.

Design Method: Usability Testing

Testing: For the medium-fidelity prototype test, I (Greg) did a Usability Test on a 12 year old male with some music experience in that he practices playing the saxophone in his junior high band. The participant would spend around an hour with our music website application testing the sound quality and other features of the application. Below are the questions I asked the participant along with their responses:

Questions:

1. Did you find this application easy to use or fun to use?
2. What did you think of the sound quality of the application?
3. Was there anything confusing when using the application when trying to make your music?
4. Have you spent time using other kinds of music creating applications that may be similar to our application?
5. Did you think the UI design and UX design of the application was good?
6. Are there any improvements for the application you would like to see?
7. “How close is the sound to actual physical musical instruments?”
 - a. 1. Not close at all
 - b. 2. Somewhat not close
 - c. 3. In the middle
 - d. 4. Pretty close
 - e. 5. Very close
8. “How important is it to you knowing you can create music without disturbing your friends, family or neighbors?”
 - a. 1. Not important at all
 - b. 2. Somewhat not important
 - c. 3. In the middle

- d. 4. Kind of important
 - e. 5. Extremely Important
9. "How important is it to you knowing you can collaborate with other musicians when using this application and share the music you create with other musicians online?"
- a. 1. Not important at all
 - b. 2. Somewhat not important
 - c. 3. In the middle
 - d. 4. Kind of important
 - e. 5. Extremely Important

Answers:

1. The participant said the application was easy to use and kind of fun.
2. The participant said the quality of sound was really good.
3. The participant got confused as to how to use the Overlay page but once I walked him through what to do, he got the hang of it.
4. The participant said he once used a music application that only had piano sounds and said this application was better.
5. I had to first explain to the participant what UX design is, and what UI design is, and after that he said the look of the website was good, and how things were laid out on the website for the most part easy to understand.
6. The participant said he wanted more instruments to be added to the application. He wanted the score sheet to not move the key mapping section on the Studio page. He wanted more options on the Overlay page like with how to edit tracks. He wanted a better background on the Instrument Selection page. Wanted to show on the score sheet when a note is being held.
7. He gave this a 5. Said it sounds like an actual instrument.
8. Gave this a 1. Said he does not care about this.
9. Gave this a 3. Said it might be cool to share music he makes with friends.

Results: The participant seemed to have fun using the application. He was not much of a talker, but he seemed to understand how to make music, how to change instruments, where the downloads went when he stopped his recordings, and how to download the score he created. He got a little confused when it came time to use the Overlay page, but I guided him through it and he created some recordings and overlaid his own track. He gave scores of 5, 1, and 3, which obviously is an average of 3 so this test is not successful in that the average was below 4. However, I think watching the participant have fun with the application and getting his feedback to make the website application better for me what made this test a success.

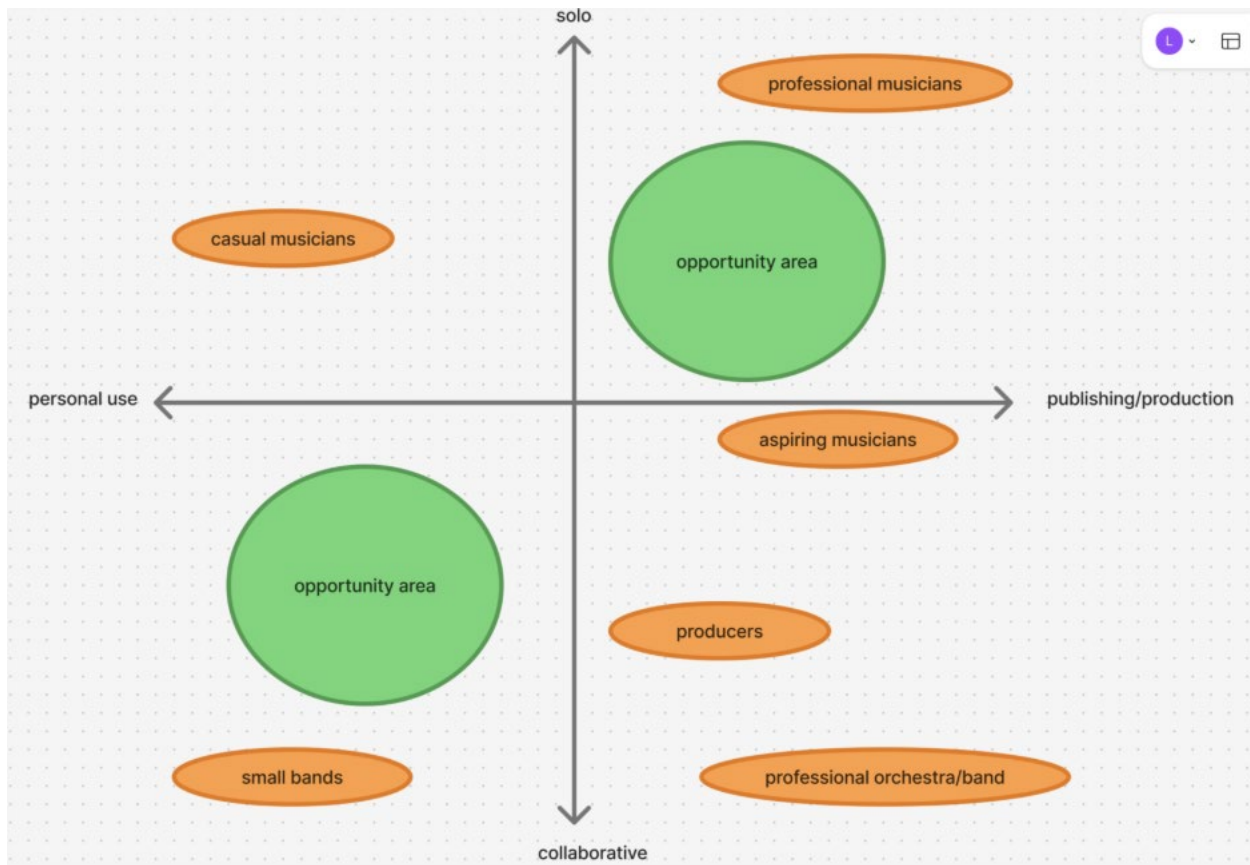
Conclusion: I think right now the Glowworm music making website application is good, but we have many improvements to make before this is a finished product that users will really love to use.

VI. Metrics

Define	Measurement	Description
Users	Number of users that have used the application.	This will be the number of users that have went to the website and made a song in the last week and month.
Activity	How much time each user spends on the application.	Will track the amount of time users are logged into the application and making music.
Learning How to Read and Write Musical Score Sheets	Will determine if users are downloading musical score sheets.	Can track when users download a musical score sheet and how much time users are on the musical score sheet page of the website.
Features of Application	Determine how users are using the application.	Find out the number each instrument sound is being used and rank them, number of times users create a song and download the song, have a share song feature on the website to track number of times songs are given to other people, and track the number of times users are taking advantage of the overlay track feature.

VII. Design Documentation

Mode 1.12 Initial Opportunity Map



Findings: Based on the map, we could focus on a PWA that supports those musicians that want to make music collaboratively on a smaller scale. Currently, working in a group would require a lot more money and investment than simply using a single tool yourself. There are also issues sharing resources among members and working in real time without professional setups.

Mode 2.10 Industry Diagnostics

Step 1: Gather information about the industry.

Prepare to assess the impact of all the Five Forces in your industry: potential entrants, substitutes, buyers, suppliers, and competitors. Review information that you may have already gathered that relate to any of these Five Forces and be prepared to do additional research if needed.

I (Greg) prepared to assess the impact of all the Five Forces in our industry of digital music production, the industry of physical musical instruments, and applications for casual music creators. I did research online to help me assess the impact of the Five Forces which are potential entrants, substitutes, buyers, suppliers, and competitors.

Step 2: Assess the impact of the Five Forces on the industry.

1. Potential entrants: Using industry information, ask who might be new entrants to the market. How high are the barriers to their entry? How vulnerable is the industry to threats from these new entrants?

New entrants to the market would be other groups in our ACMP 4260 class making a digital musical instrument Progressive Web Application like us. The barriers to their entry are very low because it does not take much time, resources, or money to make a Progressive Web Application that will act like a musical instrument that produces sound. Other groups just need to know how to work with HTML, CSS, JavaScript, and a database like AWS or Firestore. Other teams just have to put in the effort hours to complete the application. The vulnerability from these threats to the industry from these new entrants is very high. Other teams could make a better digital musical instrument application with lower latency, better UI design, and a better UX design than our application. There are also digital music companies in the industry like Chrome Music Lab, Soundtrap by Spotify, BandLab, OnlinePianist, VirtualPiano, and many more that are threats to our application. We should not forget there are music companies that make music synthesizers like Roland, Korg, Yamaha, Moog, and Behringer, Arturia, Nord (Clavia), Sequential, Teenage Engineering, and UDO Audio. There are of course companies that make actual physical musical instruments that would be a threat to our application.

2. Substitute offerings: Ask how readily customers may switch to alternative or substitute offerings. How vulnerable is the industry to the threats from such substitute offerings?

I think customers switching to alternative or substitute offerings would depend on price and the kind of sound customers are looking to hear. Music synthesizers can cost a lot of money, but there are also free music synthesizers online, so that could be a threat to our application and make us vulnerable to their substitute offerings. Music Synthesizers could produce a clearer and better sound compared to our application so customers may switch because of that. There is also the vulnerability that comes from other groups in our ACMP class that are making offerings similar to our offering. Real physical instruments are alternatives or

substitutions that customers may switch to because people may like the feel better than some digital keyboard making sound application. That is vulnerability and something we need to notice as a threat to our application.

3. Customers/buyers: Ask how much control customers have in dictating the kinds of products and services available in the industry. How much influence do customers have in pricing or other attributes of the offering?

Customers have a lot of control in dictating the kinds of products and services available in the industry. Customers also have much influence in pricing and other attributes of the offering. Music teachers in grade school, middle school, and high school have a say in the music and musical instruments they teach their students. Students in school also have a choice in what instrument they want to learn or music they want to make. If music teachers or students can get cheap or free musical instruments on Facebook Marketplace or Craigslist our progressive web application that makes music has to really stand out. Music teachers, students, musicians, and other customers will want to use our application and not real physical instruments, synthesizers, or applications similar to ours.

4. Suppliers: Ask how suppliers exert their demands. How reliant is the industry on the suppliers? How much control do suppliers have in determining the kinds of products and services the industry produces?

Since we are making a Progressive Web Application that is a digital application we really do not have to worry about physical supplies like hardware products. For us the supplier is the database that holds our application that will allow many users to make music. The database could be AWS, Firestore, Google Cloud, Azure, Supabase, PlanetScale, or many others. The only thing we have to worry about is we are adhering to the licensing agreement to the database hosting our application. We have to make sure our database is secure and only grants access to those allowed to be in the database. The supplier could demand payment if we have overages on the amount of data our application uses because of many visitors to our website using the application.

5. Competitors: Understand the nature of rivalry among competitors. Is it technology driven, price driven, or service driven? Is the rivalry consistent across competitors or variable?

I think the nature of the rivalry among competitors is technology driven, price driven and service driven. I mentioned before that there are already competitors out there like Chrome Music Lab, Soundtrap by Spotify, BandLab, OnlinePianist, and VirtualPiano. Which one customers use depends on which produces the best sound, which has the best UI design and which has the best UX design because they are all free, however Soundtrap by Spotify, OnlinePianist, and VirtualPiano you will have to pay for more options. Price is important because nobody likes to pay for something that may not produce the sound the customer wants or does not have the ease of use they may be looking for in a product. Obviously physical musical instrument makers that are our competitors are usually not free and cost money, sometimes a lot of money. Companies that are competitions to use include Yamaha, Eastman Music Company, Gibson, Fender, Vincent Bach, Jupiter, Steinway & Sons, Kawai, Saga Musical Instruments, etc. This is also where competition is service driven because people like to connect to one another and enjoy experiences together and going to Guitar Center or West Music to learn to play an instrument is

an important part of many people's lives. Also going to concerts and orchestras and symphonies is something many people enjoy and take pleasure in going to concerts, orchestras, and symphonies that musicians practice for to become the very best. There probably not a lot of people on this planet going to a Progressive Web Application music concert, orchestra or symphony. I would say the rivalry is consistent across competitors because we all want people that like music to use our product, which would be a physical musical instrument or synthesizer, or musical application, which would be the digital Progressive Web Application. I hope this all makes sense and is logical.

Step 3: Identify the organization's response mechanisms.

Mechanisms are organization's activities that can respond to these Five Forces. Examples include policies, procedures, plans, budgets, controls, and protocols. What mechanisms are in place to monitor changes and make corrections as required?

The best way our team can respond to these Five Forces is to make the application very easy to understand how to use. A nice description of what our application does and how to use it will be on our website, and this will make users want to play music on our application. I think if our application has an easy to use control scheme then users will come to our application to play music. Our application will be the same reason people like certain video games and playing physical musical instruments, and the reason is with practice players can get better and better. Our application will also have the ability to automatically save the notes users have played and our application will create a musical score sheet automatically for users. This can help users understand how to read music as well as write music. Our application was going to use the shift and spacebar keys to change the pitch and length of musical notes played by users. That could differentiate us from other applications out there right now. Our application would also allow users to select from which instrument they want to make sound from, which for now we will have the guitar, the piano, the violin, and the trumpet.

Step 4: Discuss the findings and look for insights.

Put together all the findings from previous steps in a presentational form. Discuss this as a team. What are the opportunities for your organization to play a new role in the industry?

We feel as a team that even though we have competition from other teams in our ACMP 4260 course, and from other Progressive Web Applications made from large and powerful companies our application will be a different and better musical experience for people just entering into the digital music making world, novice musician, veterans in digital music, virtuoso musicians, and everybody in between. Our UI design and UX design for our website will be so innovated, easy to navigate, aesthetically pleasing, and the sound will be identical to an actual physical musical instrument that users will not want to leave our website.

Mode 2.13 Interest Groups Discussion

Step 1: Seek out interest group forums.

- **Group 1: In Game + Browser Music Creation communities**
 - Mabinogi Official Forums (Music Section)
 - BeepBox
- **Group 2: Music Score Sharing + Collaboration Platforms**
 - Flat.io forums
 - MuseScore community forum

Step 2: What is being discussed

- **Group 1:**
 - Mabinogi players are discussing how to make music without learning MML (Music Macro Language). Some of the common questions in these threads surround how to make music/songs without any musical knowledge.
 - A thread surrounding a requested feature where players could play music live with their keyboard and have it automatically written to a score.
 - BeepBox is known for its accessibility to non-musicians. You simply click on a grid to compose music.
- **Group 2:**
 - On both Flat.io and MuseScore users ask about editing scores simultaneously with others and while Flat.io supports this natively MuseScore doesn't
 - Users on both Flat.io and Muse Score want easier ways to discover and access other people's compositions.

Step 3-4: Dive in and Capture your Findings

After diving into the forums, I've captured the findings below:

- **Finding 1:** There seems to be a frustration over having to learn MML to create music in Mabinogi than simply playing music and having the system record it.
- **Finding 2:** URL/link-based sharing seems to be the best approach to sharing compositions. BeepBox's approach to sharing music is simple and doesn't require accounts or file management.
- **Finding 3:** Multi Instrument "band" composition has demand, but a solution doesn't seem to exist so far. The closest platform may be Flat.io (come back to
- **Finding 4:** Key customization is a need, the forums discuss the need for keyboard customization

Step 5: Review and Summarize

After reviewing the findings, some of the themes that have come up are:

- The desired workflow would be to play first and have the system notate later.

- The existing tools are either too complex or too limited (For casual users' notation tools like Flat.io and MusiceScore are too intimidating)
- Composition sharing builds community although attribution mechanics are necessary, attribution features that include clear creator credit and permission controls.

Mode 4.20 Results

Step 1-2: Gather all the insight clusters generated from many methods. Normalize them into a finite list of insights.

- Current DAWs force users to go through loopholes to collaborate with other in real time
- AI tools make producing easier, but are found by some users to suck the joy out of making music
- DAWs are an alternative to expensive studio time for smaller artists
- New users often face a sharp learning curve
- The complexity of many software programs can overwhelm artists
- For users new to music production, lack of foundational knowledge makes using these platforms even more difficult
- Some users would also appreciate being able to share their creations with other member of the platform
- Artists can become reliant on presets and digital tools and feel a loss of connection with their actual skills

Step 3-4: Generate 3-10 high level design principles

- **A live call function where users can work on a project in real time**
 - Current DAWs force users to go through loopholes to collaborate with other in real time
- **An optional AI helper/co-pilot that provides resources to foundational building tools and can help make music**
 - AI tools make producing easier, but are found by some users to suck the joy out of making music
 - New users often face a sharp learning curve
 - For users new to music production, lack of foundational knowledge makes using these platforms even more difficult
- **Publishing pieces of music, beats, etc. on the platform for community use**
 - Some users would also appreciate being able to share their creations with other member of the platform
- **Options to synthesize music and to record/edit recorded music**
 - Artists can become reliant on presets and digital tools and feel a loss of connection with their actual skills
- **Clear tutorial when the app is opened**
 - New users often face a sharp learning curve
- **Free or reasonable pricing for different tiers to remain accessible**
 - DAWs are an alternative to expensive studio time for smaller artists

Mode 5.6 Concept-Generating Matrix

Step 1. Select two set of factors to build the matrix

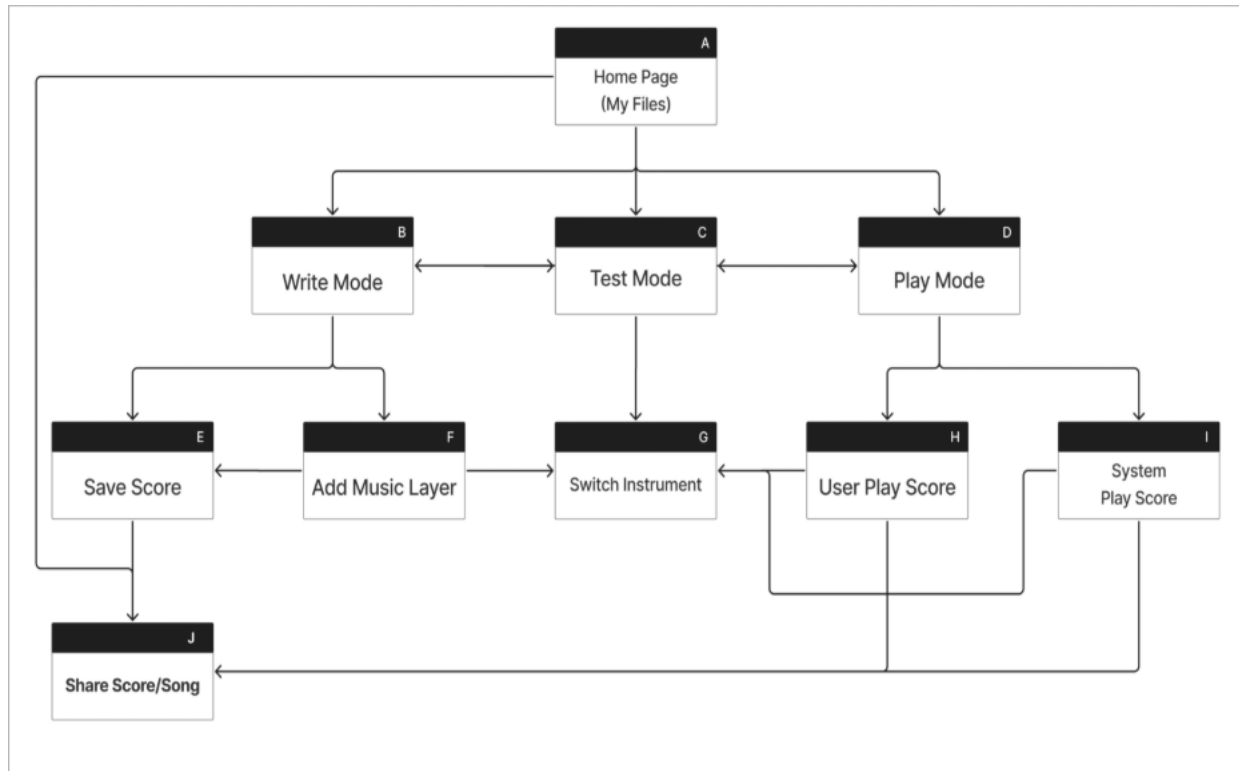
- Rows axis 1: user types (the newcomer, the professional producer)
- Columns axis 2 – design principles: synth + record/edit, publish & share, real-time co-creation

Step 2. Populate the Matrix Cells with Concepts

	Synth + record/ edit	Publish & Share	Real - time co-creation
Beginner (Newcomer)	Play-along mode Record over pre-built stems	Beginner showcase Safe feed for first creations	Guided jam room Tutorial-led live session for two beginners
The Producer (Professional)	Modular channel strip Build custom signal chains from scratch	Stems library Publish stems for community remixing	Async collab thread Timestamped comments on shared projects

Mode 6.6 Solution Diagramming

This is a flowchart/process map to show the process of the user using the PWA to make a song.



- When the user opens app, they start on the home page (assume they are logged in), which contains all their music files. **(A)**
- When a user goes to create a new score, or selects an existing music file, they automatically start in write mode. **(B)**
- The user can press configured keys on their keyboard to write the score automatically. Afterwards, they can save the current score with the current instrument **(E)**, or they can add a new music layer that will play on top of the previous score(s). **(F)**
- From write mode OR play mode, the user can also switch the instrument associated with a given score layer. **(G)**
- When user is done with song, they can save their music file and share it **(J)** for others to play and/or edit.
- User can switch to test mode to experiment with different instruments without writing anything **(C)**, or switch to play mode **(D)**, where they can manually play their score with their keyboard **(H)** or have the system play it. **(I)**

Mode 7.2 Platform Plan

STEP 1: Discuss key attributes of successful platforms.

- Manuals and tutorials including built in lessons
- Free downloads of public user generated content
- Browsing of user generated content
- Collaborative studio
- Virtual instruments
- Connect instruments to record sound
- Podcast support (transcribing, integration with Spotify)
- Education features
 - Integration with major LMS systems
 - Lesson plans, assignments, etc.

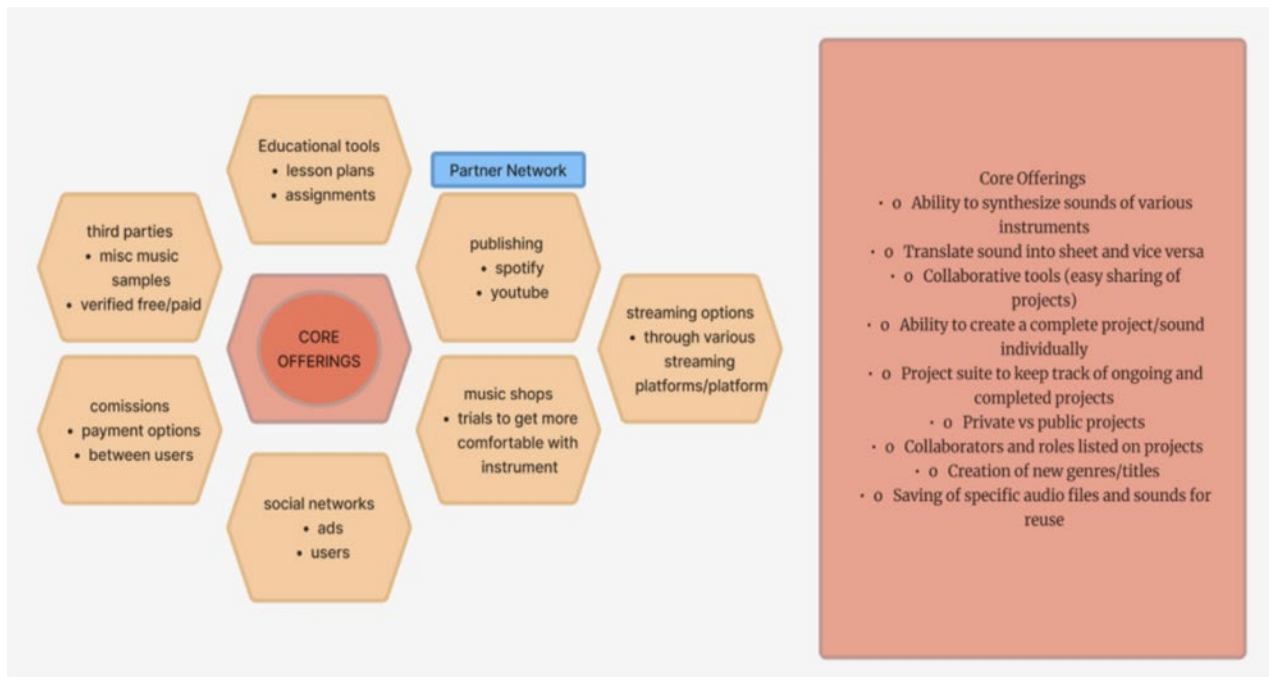
STEP 2-3: Identify solutions/initiatives that have the potential to be a platform over time. Use the four platform definitions to review solutions.

- **Core with Options**
 - Ability to synthesize sounds of various instruments
 - Translate sound into sheet and vice versa
 - Collaborative tools (easy sharing of projects)
 - Ability to create a complete project/sound individually
- **Sticky Users**
 - Project suite to keep track of ongoing and completed projects
 - Private vs public projects
 - Collaborators and roles listed on projects
 - Creation of new genres/titles
 - Saving of specific audio files and sounds for reuse
- **Distributed Owners**
 - Users
 - Integrated platforms (Spotify, Canvas)
- **Open Partners**
 - Companies that provide free or paid samples

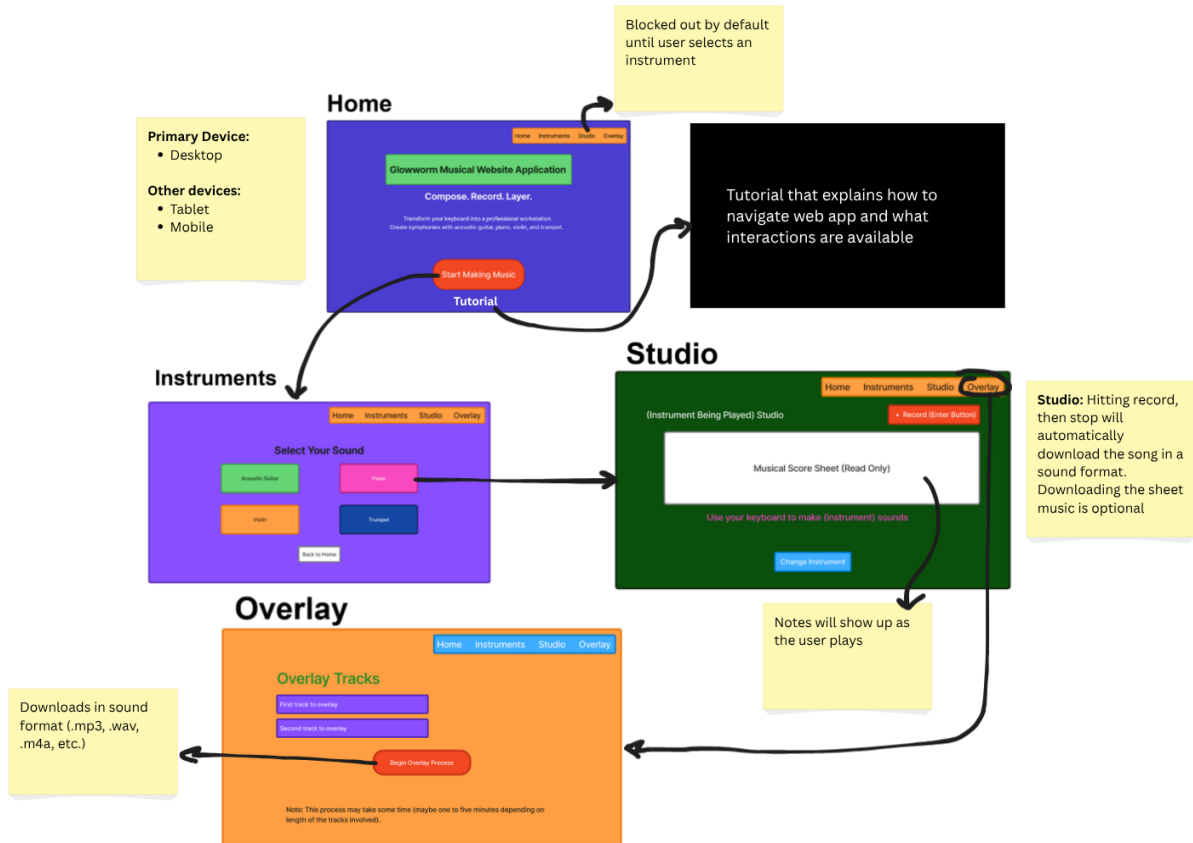
STEP 4: Use the key platform attributes to review solutions.

- **Must Have Attributes**
 - Accessibility
 - Connectivity
 - Collaboration
 - Sharing
 - Ease of use/user friendly interface
- **Nice to Have Attributes**
 - Scalability
 - Modularity
 - Extensions

STEP 5: Create a plan for platform implementation.



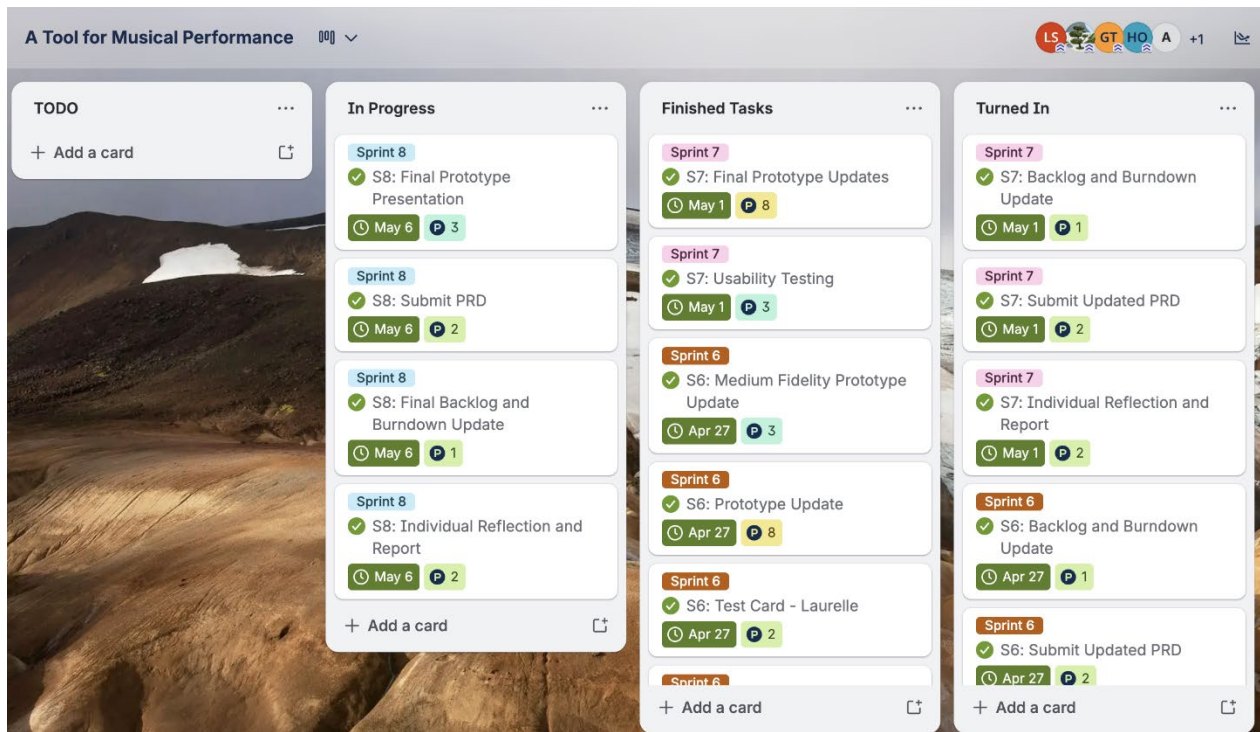
Wireframe Diagram



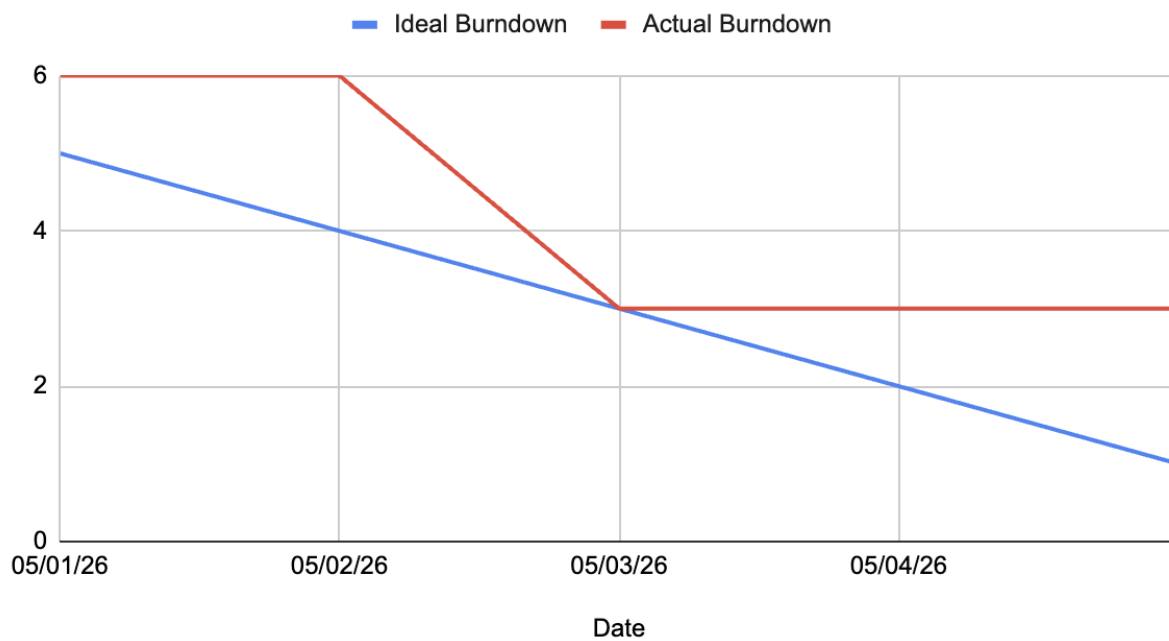
VIII. Timeline for Development

Start Date	Title	Tasks	Estimated Total Effort Hours	Actual Total Effort Hours
3/11/26	S0: Sprint Planning	- Outline draft PRD - List tasks for each sprint - Create a Kanban and burndown	3	3
3/23/26	S1: Discovery	- Execute three design methods corresponding to design Modes 1-3 - Craft an intent statement - Update Backlog and Burndown - Complete S1 personal reflection - Update draft PRD	9.75-12.75	14.75
3/25/26	S2: Definition	- Select and execute one design method corresponding to design <u>Mode 4</u> - Create <u>9 personas</u> - Update Backlog and Burndown - Complete S2 personal reflection - Update draft PRD	9.75-12.75	11
3/30/26	S3: Development and Delivery	- Select and execute at least three design methods corresponding to design Modes 5-7 - Present an idea pitch during class 2 - Create Test Cards - Use design methods to define test procedures - Define measurable Key Performance Indicators (KPIs) - Update Backlog and Burndown - Complete S3 personal reflection - Update draft PRD	10.25-13.75	13
4/6/26	S4: Low-Fidelity Prototype Testing	- Create a low-fidelity prototype - Test the low-fidelity prototype for using a selected design method and fill out at least one test card - Write a report on your testing, results, and conclusions - Update Backlog and Burndown - Complete S4 personal reflection - Update draft PRD	24-28	13.5

4/13/26	S5: Develop and Debug	• Class 1 - Present low-fidelity prototype report (slide deck recommended) • Develop a medium-fidelity prototype • Update Backlog and Burndown • Complete S5 personal reflection • Update draft PRD	16-20	21.5
4/20/26	S6: Medium-Fidelity Prototype Testing	• Create a medium-fidelity prototype • Test the medium-fidelity prototype for using a selected design method and fill out a test cards • Write a report on your testing, results, and conclusions • Update Backlog and Burndown • Complete S6 personal reflection • Update draft PRD	24-28	25
4/27/26	S7: Development and Debugging + High Fidelity Testing	• Develop final high-fidelity prototype • Test the high-fidelity prototype for using a selected design method and fill out a test cards • Write a report on testing, results, and conclusions • Update Backlog and Burndown • Complete S7 personal reflection • Update draft PRD	24-28	12.5
4/29/26	S8: Final Presentation	• Finalize high-fidelity prototype • Update Backlog and Burndown • Complete S8 personal reflection • Finalize PRD • Create and present a Final Presentation	24-28	8.5

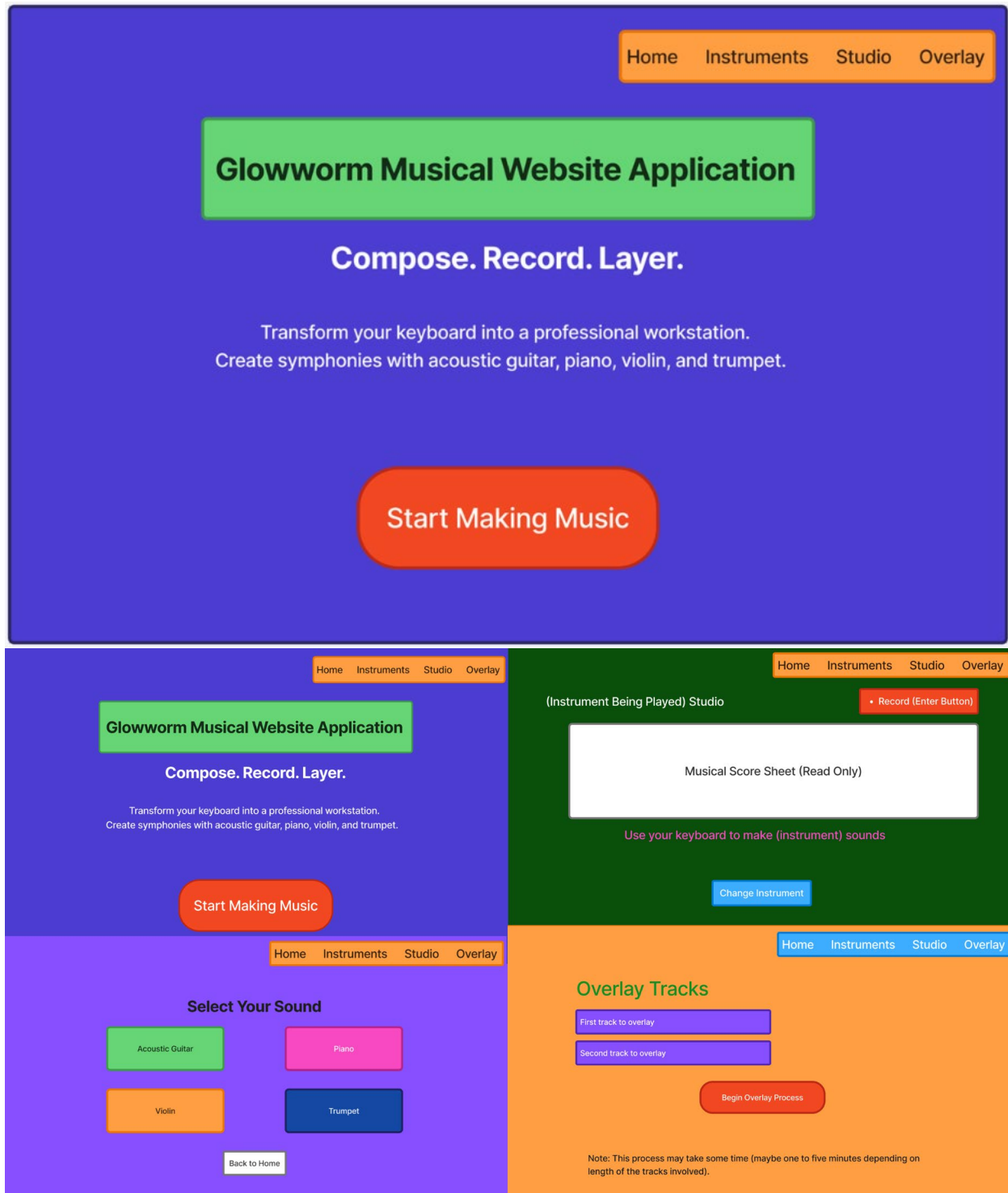


S8: Burndown Chart



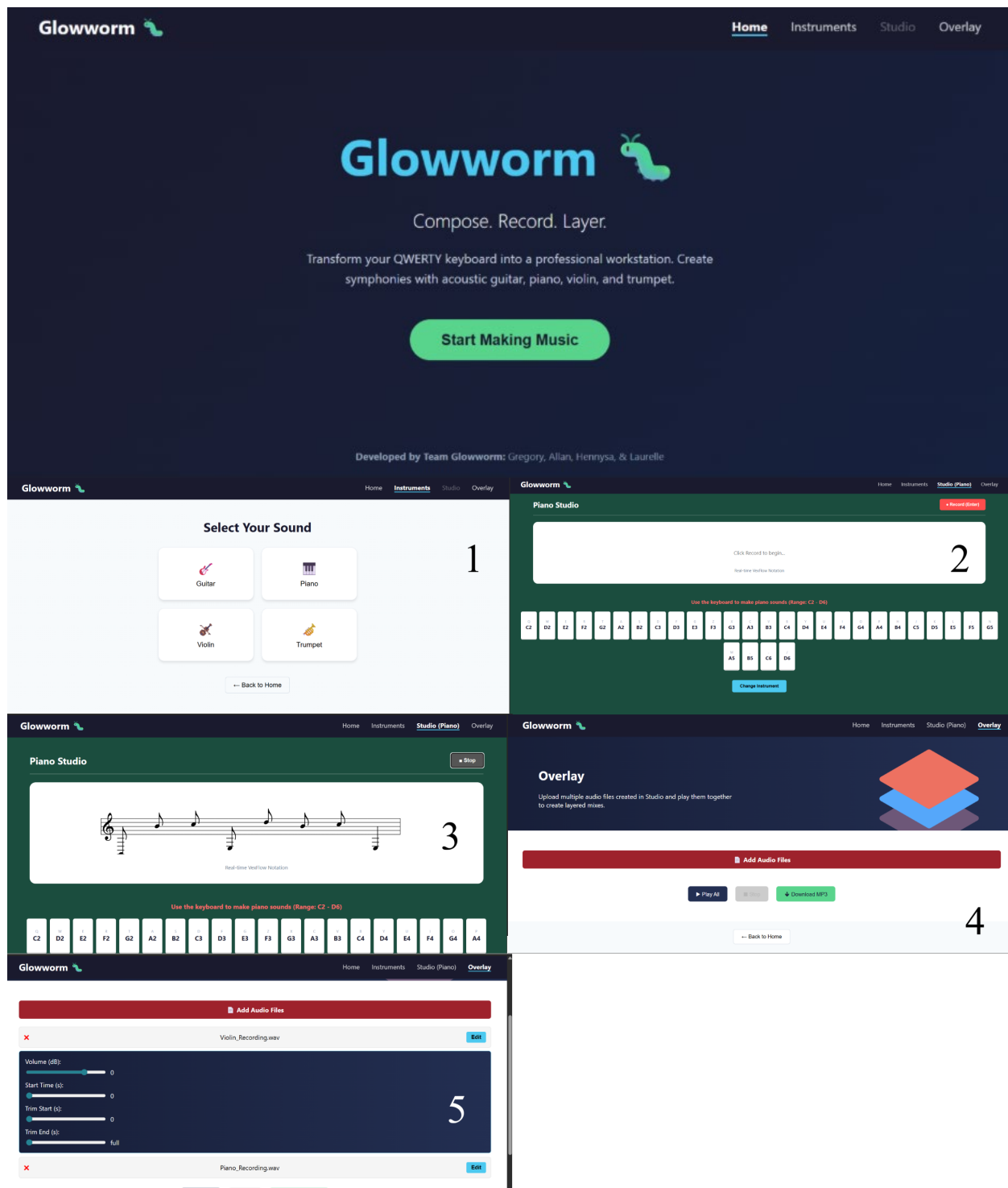
IX. Product Documentation

Low-Fidelity Prototype



Home screen of low-fidelity prototype. Users can click “Instruments” or “Start Making Music” to select their instrument of choice, then click “Studio” to start playing and recording a track. In “Overlay,” users can then layer their tracks to create a “band” or song.

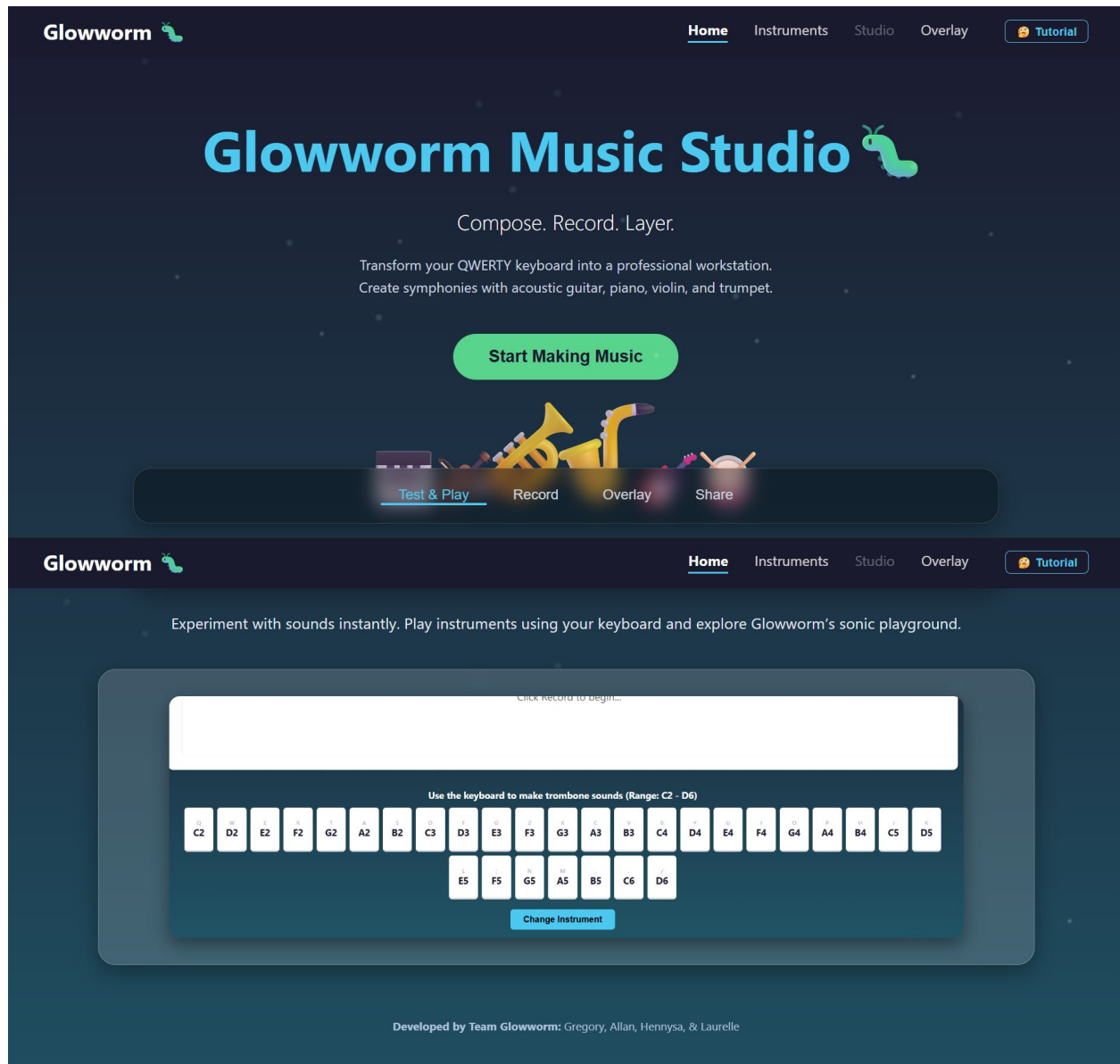
Medium-Fidelity Prototype



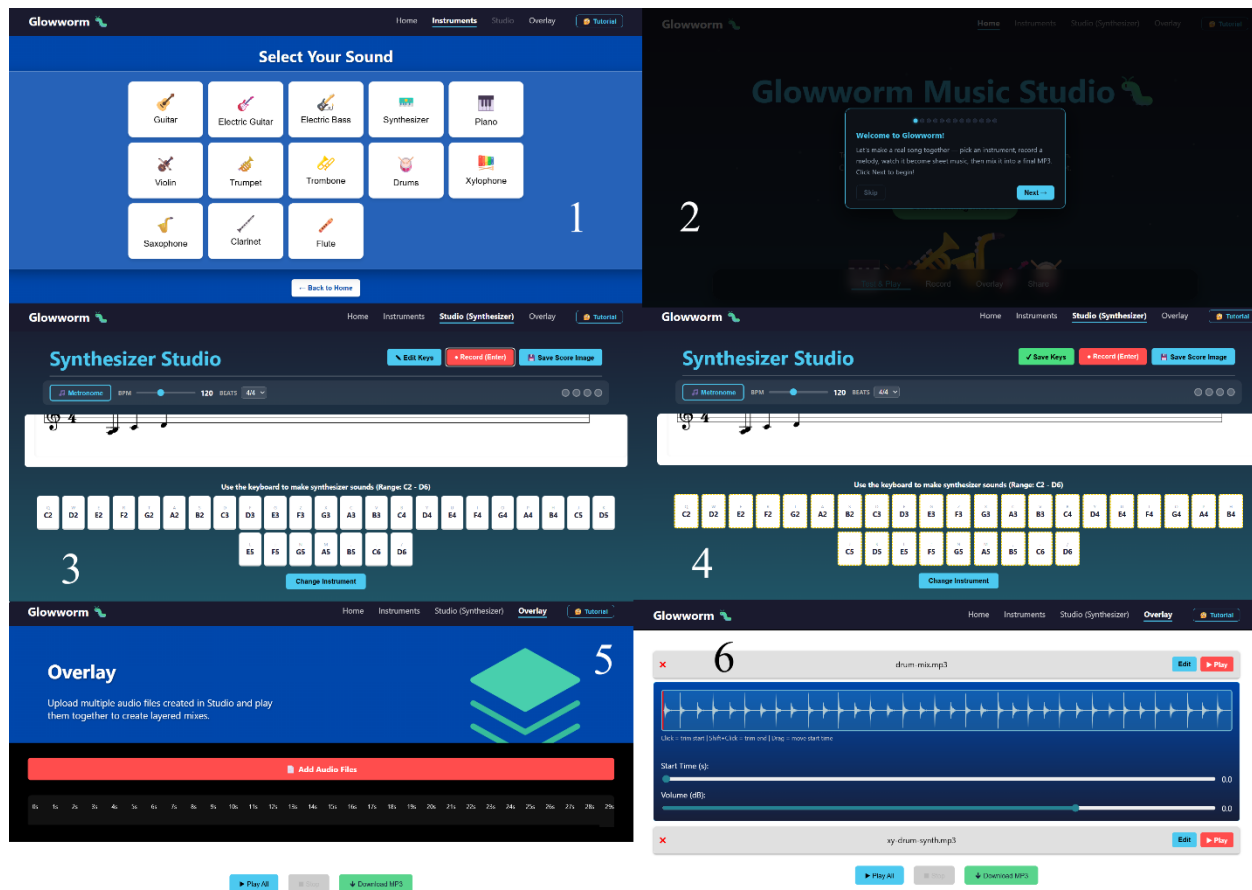
Home screen of medium-fidelity prototype: 1. Users can click “Instruments” or “Start Making Music” to select their instrument of choice, 2. then click “Studio” to start playing and recording a track. Typing with the keyboard will 3. cause notes to start showing on the screen. After pressing “Stop,” the track will automatically download, and the user can optionally download the created

sheet music. **4.** In “Overlay,” users can then layer their tracks to create a “band” or song. **5.** Optionally, users can mess around with the volume, timing, or trim the beginning or end of certain tracks.

High-Fidelity Prototype



URL: https://ysabum.github.io/archive/UX_Design/glowworm_music_studio/



Home screen(s) of high-fidelity prototype: 1. Users can click “Instruments” or “Start Making Music” to select their instrument of choice, 3. then click “Studio” to start playing and recording a track. Typing with the keyboard will cause notes to start showing on the screen. After pressing “Stop,” the track will automatically download, and the user can optionally download the created sheet music. There is also a customizable metronome that can be used to keep time while recording (the metronome will not be heard in the recorded track). 4. Users can also edit the default key bindings. These will persist over page refreshes. 5. In “Overlay,” users can then layer their tracks to create a “band” or song. 6. Optionally, users can mess around with the volume, timing, or trim the beginning or end of certain tracks. While editing in Overlay, users can also listen to the entire layered song or each song individually.

2. There is also a skippable tutorial that takes new users through the web application and explains its main functionalities.

X. Student Learning Reflections

Sprint 1 Reflection

1. **What do you think went best (either in outcome or process) during this sprint?**
 - a. **Allan:** I think the task distribution went well and our tasks were very clear.
 - b. **Greg:** I feel what went best was the way every task was distributed in the group. We are all working as a team to think of the idea for a great progressive web application and then create the application.
 - c. **Henny:** I think since we properly divided up our tasks in Sprint 0 and selected the methods/modes we would execute for this sprint and other future sprints, completing our assigned tasks (for me at least) went smoothly.
 - d. **Laurelle:** I think our planning and splitting up of tasks worked well in this sprint. It is helpful having everyone do some parts individually as we don't have much time to all meet this early in the week.
2. **What do you think went worst (either in outcome or process) during this sprint?**
 - a. **Allan:** On my part I missed the first class of the week, so it took a bit to catch up with my group . Overall, the sprint outcome went well!
 - b. **Greg:** I think during this sprint nothing went too badly. It took a lot of effort hours to complete the 2.10 Mode, but it was not that bad. Everything seems to be coming along nicely.
 - c. **Henny:** For me, I don't think anything went poorly for this sprint other than needing to spend additional time brainstorming what our project idea would be since we did not do beforehand.
 - d. **Laurelle:** For me, the burndown chart went a bit poorly because it took me much more time than expected to and I feel it could've been quicker and more efficient.
3. **How do you think you personally contributed to making your group's sprint go better?**
 - a. **Allan:** I personally contributed by completing my assigned tasks.
 - b. **Greg:** I helped with coming up for the idea that our progressive web application will be. I also worked on Mode 2.10 Industry Diagnostics.
 - c. **Henny:** For sprint 0, I was responsible for dividing the tasks up, and I also helped brainstorm what modes we should execute for future sprints. I think this all contributed to making completing sprint 1 go well.
 - d. **Laurelle:** I would say that I did contribute to making the sprint go better by completing my assigned tasks on time.
4. **What was the hardest part of working on a team during this sprint?**
 - a. **Allan:** I wasn't present on Monday so unfortunately, I wasn't able to contribute as much as I wanted to.
 - b. **Greg:** The hardest part was coming up with an idea for the progressive web application we will be creating as a team.
 - c. **Henny:** I think the hardest part of working on a team for this sprint was, when we were brainstorming project ideas, needing to figure out how to put our ideas together.

- d. **Laurelle:** The hardest part of working on a team during this sprint was making sure that any tools we use are accessible to other members so that no one is left out of viewing the team's progress.
5. **How could you make the experience of working on a team go more smoothly for you during the next sprint?**
- a. **Allan:** I am sure being present would be helpful and reviewing the sprints/tasks asks beforehand would help prime everyone for a productive workday in the class.
 - b. **Greg:** Do a review before class to see what we are going to be working on. That way we are all prepared with ideas we may have, questions we may have, or comments we may have about the project we are working on.
 - c. **Henny:** I think reviewing the task schedule from Sprint 0 as well as looking at the class agenda so we can individually come prepared to class or a meeting will help in making the experience of working on a team go more smoothly.
 - d. **Laurelle:** During the next sprint I think it would be helpful to more quickly decide what we need to work on collaboratively and what we will be splitting up to work on.

Sprint 2 Reflection

1. **What do you think went best (either in outcome or process) during this sprint?**
 - a. **Allan:** I think what went well was the completion of our tasks, these gave us the foundation for our closing meeting.
 - b. **Greg:** I think how the work was spread out between all of the team members went well. I also think the personas I came up with were good, and the gains, pains, and jobs for each persona were done well.
 - c. **Henny:** Like the last sprint, I think actually completing our individual tasks went well as we did a good job splitting up the tasks before the start date for sprint 2.
 - d. **Laurelle:** I think what went best was our planning and spreading out of work in this sprint.
2. **What do you think went worst (either in outcome or process) during this sprint?**
 - a. **Allan:** I don't think anything went badly or worse. It's been a smooth productive ride so far.
 - b. **Greg:** I do not think anything went badly or worst for this sprint.
 - c. **Henny:** I don't think anything went poorly for this sprint. I mostly had difficulty trying to brainstorm pains, gains, and jobs for my personas.
 - d. **Laurelle:** I think this sprint went pretty smoothly and we were able to follow our plan.
3. **How do you think you personally contributed to making your group's sprint go better?**
 - a. **Allan:** I think doing my part of the personas was necessary for the group sprint go better since it was part of the foundation of sprint 2
 - b. **Greg:** I think coming up with the personas and gains, pains, and jobs for each persona was a nice contribution to this sprint.
 - c. **Henny:** I was mostly responsible for coming up with my 3 personas for the 41-65 age group. I think doing this was helpful in our final task of ranking our personas in importance.
 - d. **Laurelle:** I think the insights and features will help with the details of our final design.
4. **What was the hardest part of working on a team during this sprint?**
 - a. **Allan:** compiling and navigating our ideas takes a bit of time but that's what I found to be the hardest part during this sprint
 - b. **Greg:** Thinking of the different pains, gains, and jobs for each persona took a lot of time.
 - c. **Henny:** As I said for question 2, I mostly had difficulty trying to brainstorm pains, gains, and jobs for my personas.
 - d. **Laurelle:** The hardest part was still checking in with team members even if your task is individual.
5. **How could you make the experience of working on a team go more smoothly for you during the next sprint?**
 - a. **Allan:** Preparing and reviewing the next sprint helps navigate it a bit smoother.

- b. **Greg:** Looking ahead at what we have to work on next and trying to come up with ideas when I have free time could help make the next sprint go better.
- c. **Henny:** I think reviewing the task schedule and class agenda, as well as trying to complete work ahead of time if possible, will help things go more smoothly for the next sprint.
- d. **Laurelle:** I think what we did for this sprint went well and I would like to keep the same routine of splitting up tasks and meeting up to share progress.

Sprint 3 Reflection

1. **What do you think went best (either in outcome or process) during this sprint?**
 - a. **Allan:** I think sprint was organized well
 - b. **Greg:** I think what went best was coming up with the tests the Test Cards. I also think the questions and answers went well.
 - c. **Henny:** Executing Mode 6.6 went fairly well.
 - d. **Laurelle:** I would say our separation of tasks and organization went well for this sprint.
2. **What do you think went worst (either in outcome or process) during this sprint?**
 - a. **Allan:** I don't think anything was worse this sprint
 - b. **Greg:** Finding a Test Card I could write text on or an editor for a PDF file took longer than I expected and made me a little frustrated.
 - c. **Henny:** For me, I don't think anything went too poorly during this sprint.
 - d. **Laurelle:** I don't think there were any problem areas for this sprint.
3. **How do you think you personally contributed to making your group's sprint go better?**
 - a. **Allan:** The completion of mode 5.6 opened some new ideas and angles on the subject at hand (music production)
 - b. **Greg:** I think making the Test Cards and Key Performance Indicators table is a good contribution to our group's sprint.
 - c. **Henny:** I think making a flowchart of how the different pages of our application are connected and how the user would interact with the application was helpful in making our solution to the design brief clearer.
 - d. **Laurelle:** I contributed by completing my assigned task so that we can continue developing our project.
4. **What was the hardest part of working on a team during this sprint?**
 - a. **Allan:** There wasn't really a hard part with working on a team on this sprint, everyone communicates well.
 - b. **Greg:** I do not feel there was ever a hardest part of working on a team during this sprint. We all did our part when we worked on this sprint and things are going well.
 - c. **Henny:** Some of our task assignments were changed since they were assigned randomly without knowing what the actual task would entail. So, I would say the hardest part was just figuring out what each team member needed to do for this sprint after reassignment.
 - d. **Laurelle:** For this sprint, the hardest part was coming up with what the platform plan should look like because there was no specific example in the book.
5. **How could you make the experience of working on a team go more smoothly for you during the next sprint?**
 - a. **Allan:** Always good to prepare before class so in class work is smooth sailing.
 - b. **Greg:** Need to better prepare for the work I will do for the next sprint by looking over my tasks and not wasting time on things that are not that important.

- c. **Henny:** I should probably do task reassignments and message the team members before class just so there is less time spent discussing this and more time working on our tasks.
- d. **Laurelle:** I think we've figured out how to plan effectively to have a smooth sprint.

Sprint 4 Reflection

1. **What do you think went best (either in outcome or process) during this sprint?**
 - a. **Allan:** I think what went best was our testing and task distribution during this sprint.
 - b. **Greg:** I think what went best was coming up with the questions and doing the usability testing.
 - c. **Henny:** Since the testing cards needed to be modified, having the low-fidelity prototype already completed by the start of this sprint made it easier for that to be done.
 - d. **Laurelle:** I think what went best was the testing. I was able to get some really good feedback and ideas from my test.
2. **What do you think went worst (either in outcome or process) during this sprint?**
 - a. **Allan:** Executing the test was based on the availability of my participant, which took until Sunday to test.
 - b. **Greg:** Finding time to do the usability tests was kind of difficult because people are busy doing other things in their lives and finding free time is hard to do.
 - c. **Henny:** I think what went poorly was figuring out how to conduct the testing (i.e., choosing a design method, who to test, etc.)
 - d. **Laurelle:** The only thing I could think of was choosing what design method to use with our tests. They were all a little different and would have been difficult experiences to provide our users. It took a while to find the right one.
3. **How do you think you personally contributed to making your group's sprint go better?**
 - a. **Allan:** I think my testing results contributed to the groups insights which was useful during our sprint
 - b. **Greg:** I think doing these usability tests and writing the reports is a good contribution to our group's sprint.
 - c. **Henny:** Doing the usability test was helpful in figuring out what changes need to be done for medium-fidelity prototype.
 - d. **Laurelle:** I think I contributed by completing my portion of the tasks.
4. **What was the hardest part of working on a team during this sprint?**
 - a. **Allan:** There weren't really any hard parts of working on a team during this sprint.
 - b. **Greg:** While working on a team during this sprint, I never felt there was a hardest part, and nothing ever felt difficult to get finished. We all did our part when we worked on this sprint, and things are going well.
 - c. **Henny:** I think figuring out what test (since we had 4 test cards developed from sprint 3) everyone was doing and what changes needed to be made to the original test questions was the hardest part during this sprint.
 - d. **Laurelle:** I think the hardest part specifically with the testing was agreeing on questions so that it would be consistent.
5. **How could you make the experience of working on a team go more smoothly for you during the next sprint?**
 - a. **Allan:** I like preparing by reviewing my work/tasks if needed.

- b. **Greg:** It is always good to prepare for the work I will do for the next sprint by looking over my tasks and not wasting time on things that are not that important.
- c. **Henny:** As usual, making sure to review the task agenda for the day and informing team members of any changes before class will be helpful in making things go smoothly for the next sprint.
- d. **Laurelle:** I think we just need to keep doing what we are doing and keep up with communication.

Sprint 5 Reflection

1. **What do you think went best (either in outcome or process) during this sprint?**
 - a. **Allan:** I think the development of the pwa from my peers went well
 - b. **Greg:** The fact that the website is working and looks good is something I think went the best during this sprint.
 - c. **Henny:** I think working on adding new pages to the medium prototype went well since Greg got us started a few weeks ago.
 - d. **Laurelle:** I think the medium fidelity prototype came out really well for this sprint.
2. **What do you think went worst (either in outcome or process) during this sprint?**
 - a. **Allan:** I couldn't really say, I wish I had contributed more in this sprint
 - b. **Greg:** It took a long time to get things to work correctly on the music making website application. For instance, getting the musical score sheet to put the notes that users play correctly on the score sheet was a very difficult task.
 - c. **Henny:** While maybe too advanced for the medium prototype, testing and implementing functionality for some of the pages was frustrating.
 - d. **Laurelle:** For me, I didn't feel like I helped much with finalizing or creating the medium fidelity prototype during this sprint.
3. **How do you think you personally contributed to making your group's sprint go better?**
 - a. **Allan:** I could've done more to contribute to my groups sprint.
 - b. **Greg:** I think I personally contributed to making our group's sprint go better by having the website application up and running correctly.
 - c. **Henny:** I created the Overlay page, which finalized all the pages we wanted to make for this app, so hopefully implementing any other changes will now go smoothly.
 - d. **Laurelle:** I did not contribute to the actual prototype during this sprint.
4. **What was the hardest part of working on a team during this sprint?**
 - a. **Allan:** I don't think there was a hardest part
 - b. **Greg:** There was no "hardest" part of working on a team during this sprint. I feel that our members are all working hard and contributing to this music making PWA.
 - c. **Henny:** I don't think any part of working on a team was hard, for this sprint at least.
 - d. **Laurelle:** I would say that this sprint went really well.
5. **How could you make the experience of working on a team go more smoothly for you during the next sprint?**
 - a. **Allan:** Making sure to review the PRD and distribute our tasks as usual
 - b. **Greg:** Make sure to review what is on the agenda and keep good communication with team members.
 - c. **Henny:** As usual, making sure to review the task agenda for the day and informing team members of any changes before class will be helpful in making things go smoothly for the next sprint.
 - d. **Laurelle:** I think spending more time separating tasks and adapting the agenda for our current progress will help a lot.

Sprint 6 Reflection

1. **What do you think went best (either in outcome or process) during this sprint?**
 - a. **Allan:** I think what went best was the testing since it was a lot clearer on what to do compared to the last time.
 - b. **Greg:** I think what went best was doing the Usability test and seeing what changes needed to be made to the Glowworm music making website application to make it better.
 - c. **Henny:** Doing the testing was easy this time around since we know what modifications need to be made to the test cards. Also, our prototype is essentially working, which meant we had more things to test.
 - d. **Laurelle:** I think the testing really helped us come up with ideas for improvements for the app.
2. **What do you think went worst (either in outcome or process) during this sprint?**
 - a. **Allan:** I don't think anything went worse for this sprint
 - b. **Greg:** I think what went worst was again finding a time to do the usability test because people always have things going on in their lives, so finding free time is difficult. Another worst was trying to implement the changes to the website application because there are sometimes errors that need to be corrected, and it can be time consuming finding out why things are not working properly.
 - c. **Henny:** I don't think anything went poorly for this sprint.
 - d. **Laurelle:** I wouldn't say there was anything rough about this sprint.
3. **How do you think you personally contributed to making your group's sprint go better?**
 - a. **Allan:** I added the tutorial for the application which helps solve some problems with app usability.
 - b. **Greg:** I did a test to see how the Glowworm music making website application could be made better is one way I personally contributed to making our group's sprint go better. I also think working on the changes testers wanted to see happen is another way I personally contributed to making the group's sprint go better.
 - c. **Henny:** I helped choose what test cards need to be done for this sprint, and I am working on the Overlay page for our application.
 - d. **Laurelle:** I completed the testing and got some really good feedback on the app from someone who had issues working with it.
4. **What was the hardest part of working on a team during this sprint?**
 - a. **Allan:** I don't think there was a hardest part
 - b. **Greg:** I am just going to repeat that I never felt there was a hardest part while working on a team during this sprint, and nothing ever felt difficult to get finished. We all did our part when we worked on this sprint and things are going well.
 - c. **Henny:** I don't think anything was the hardest part of working on a team for this sprint.
 - d. **Laurelle:** I think this sprint went very smoothly.

5. **How could you make the experience of working on a team go more smoothly for you during the next sprint?**
- a. **Allan:** Communicating with my team and preparing material before classes/meetings.
 - b. **Greg:** Like always, it is good to prepare for the work I will do in the coming days for the next sprint by looking over my tasks and not wasting time on things that are not that important.
 - c. **Henny:** I think just keeping up with communication and making sure everyone knows what they need to do will help in things going smoothly for the next sprint.
 - d. **Laurelle:** Looking over the week's tasks before so class so we can hit the ground running.

Sprint 7 Reflection

1. **What do you think went best (either in outcome or process) during this sprint?**
 - a. **Allan:** I think what went best was the seamless transition from a medium to high fidelity prototype.
 - b. **Greg:** I think what went best during this sprint is that our Glowworm music website application is almost completely finished.
 - c. **Henny:** Working on the final high-fidelity prototype went smoothly for this sprint since we were mostly done at this point, save for some UI changes.
 - d. **Laurelle:** I think we divided up tasks very well for this sprint.
2. **What do you think went worst (either in outcome or process) during this sprint?**
 - a. **Allan:** I think nothing went worst
 - b. **Greg:** I think what went worst is that there are just a couple of little things that need to be corrected for the Glowworm music website application to be done.
 - c. **Henny:** I don't think anything went poorly during this sprint.
 - d. **Laurelle:** I wouldn't say there was a 'worst' part.
3. **How do you think you personally contributed to making your group's sprint go better?**
 - a. **Allan:** I added a feature to the high-fidelity prototype which added some more functionality helping with the testing phase
 - b. **Greg:** I personally did not contribute much to this sprint. All I did was test out the application to see if it was working like it should.
 - c. **Henny:** I finished the UI and fixed some things with the web application, which hopefully meant testing would go better.
 - d. **Laurelle:** I contributed by doing a test with our updated prototype to get further feedback.
4. **What was the hardest part of working on a team during this sprint?**
 - a. **Allan:** I don't think there was a hardest part
 - b. **Greg:** I never felt there was a hardest part while working on a team during this sprint, and nothing ever felt difficult to get finished. We all did our part when we worked on this sprint, and things are going well.
 - c. **Henny:** I don't think anything was the hardest part of working on a team for this sprint.
 - d. **Laurelle:** We worked well together on this sprint.
5. **How could you make the experience of working on a team go more smoothly for you during the next sprint?**
 - a. **Allan:** I think preparing and understanding our tasks goes a long way, practically what we have been doing.
 - b. **Greg:** Like always, it is good to prepare for the work I will do in the coming days for the next sprint by looking over my tasks and not wasting time on things that are not that important.
 - c. **Henny:** I think we should just keep doing what we've always been doing.
 - d. **Laurelle:** I think understand what the objective is and splitting up tasks has helped.

Sprint 8 Reflection

1. **What do you think went best (either in outcome or process) in your project?**
 - a. **Allan:** I think our communication and collaborations methods helped a whole lot in the construction of our project.
 - b. **Greg:** I think what went best in our project was we each contributed something to the Glowworm website application. The communication was good between each of us for what we had to do for this project, and we always showed up to meetings when we had work that needed to be accomplished for a due date.
 - c. **Henny:** Overall, I think we had great teamwork, communication, and division of tasks, so we were able to get our final prototype done earlier than expected.
 - d. **Laurelle:** I think we worked well as a team and that we did very well with communication and keeping up with sprints.
2. **What do you think went worst (either in outcome or process) in your project?**
 - a. **Allan:** I don't think anything went worst in our project.
 - b. **Greg:** I really cannot think of anything that went worst. There were times when the application was not working correctly but all it took was troubleshooting what was wrong in the code and either fixing or updating the code.
 - c. **Henny:** I really don't think anything went worst in our project. If I really had to pick something, I would say maybe during the first two sprints (0 and 1), we were a little slow on deciding what our initial design would be?
 - d. **Laurelle:** I think we took some time to figure out how we would go about the sprints and split up work in the beginning.
3. **How do you think you personally contributed to making your group's project better?**
 - a. **Allan:** I think my brainstorming efforts and perspective could have contributed with making our project better, I added a feature that I thought would enhance the user experience alongside the walk-through tutorial.
 - b. **Greg:** I think I personally contributed to making our group's project better by beginning the development of the Glowworm music making website application. I helped with making the test cards, coming up with some personas, the low fidelity prototype, and I did some tests.
 - c. **Henny:** I guess I would say my main contribution was creating the necessary documents (e.g., reflection, PRD, any PowerPoints) for all the sprints, which may or may not have been helpful in informing my teammates what needed to be done for a given sprint.
 - d. **Laurelle:** I think I contributed to the project mainly through testing and design methods on the prototype as well as in the discovery phase.
4. **What was the hardest part of working on a team?**
 - a. **Allan:** There wasn't a hardest part working on the team
 - b. **Greg:** I never felt there was a hardest part while working on a team for this project, and nothing ever felt difficult to get finished. We all did our part in making our

Glowworm music making website application and doing the necessary documentation.

- c. **Henny:** For our team, there wasn't really a hardest part. But in general, based on my past experiences with other group projects, keeping good and frequent communication with each other as well as turning deliverables in on time were some of the hardest parts of working on a team.
 - d. **Laurelle:** I think the hardest part was figuring out how to split tasks for each sprint evenly week after week.
5. **How could you make the experience of working on a team go more smoothly for you in the future?**
- a. **Allan:** From what I learned working with my team, strong communication and collaboration channels go a long way. We were really efficient with our project.
 - b. **Greg:** I cannot think of anything that could make the experience of working on a team go more smoothly for me in the future because everything went very well working on this project with the people on my team.
 - c. **Henny:** In addition to what I mentioned for the previous question, making sure that tasks are assigned equally (if possible) and to whoever best suits those tasks are helpful in making the experience of working on a team go smoothly.
 - d. **Laurelle:** I learned a lot about what it takes to have a good experience in a team and that communication, even small updates, are very helpful.
6. **What is one piece of advice you would give to next year's students?**
- a. **Allan:** One piece of advice is to really get to know the problem and dive into the domain it covers, helps a whole lot when you have your brainstorming sessions.
 - b. **Greg:** I think one piece of advice I would give to next year's students is having people spend time with whatever you decide to make, and getting their opinions on what you are making is a great way to make that something better.
 - c. **Henny:** Try to plan ahead and complete things early, whether it's with homework or your project, since it's easy for work to pile up if you're not careful.
 - d. **Laurelle:** I would tell them to take the sprint planning document seriously because it saves a lot of time later if the sprints are basically planned out.
7. **What is one thing you would change about the class for next year?**
- a. **Allan:** One thing I would have changed is the number of assignments given to us in the first half of the semester. Although informative, they were very time consuming.
 - b. **Greg:** For me the one thing I would change would be fewer days in the classroom because I live in Fremont, NE and it took like an hour to get to class each day, so fewer days spent in the classroom and maybe spent on Zoom would have benefited me especially now that gas prices are going up.
 - c. **Henny:** Overall, for the second half of the class with our projects, I really wouldn't change anything. However, I wish there were less assignments for the first half of this class since I found the amount of them (or maybe even the spacing of assignments?) a little overwhelming.

- d. **Laurelle:** I liked the more interactive assignments in the beginning of the semester, but in the later project I would remove the individual reflection as a separate assignment and turn it into a sort of comment about the sprint students can turn in.

8. **What is the most interesting thing you learned in this class?**

- a. **Allan:** I think the most interesting thing I learned in this class is the overall system of the UX design process and how much thought is put into every element in a design. I would also say the usability testing methods were fun as they were useful in extracting feedback to improve our project.
- b. **Greg:** There were a lot of things I learned in this class that were interesting, like all the knowledge from the book A Project Guide to UX Design: For User Experience Designers in the Field or in the Making 3rd Edition by Russ Unger and Carolyn Chandler. There was the Project Ecosystem chapter that helps with planning a project, the Product Definition chapter that helps with ideas for products, and the Launch and Release chapter that talks about continually making your product better. There were also a lot of testing methods that can be done on products and other things that were interesting that I learned about from the book 101 Design Methods by Vijay Kumar. Some of these include Five Human Factors, Concept Generating Map, Solution Diagramming, Platform Plan, Interest Groups Discussion, Initial Opportunity Map, Usability Test, Behavior Prototype, and many more. I also learned that there were a lot of different music applications out there and I did not think it would be possible to help make a music making website application, but my team did, and I think that is very interesting and cool.
- c. **Henny:** I think the most interesting thing I learned in this class was how many of the methods we learned from 101 Design Methods are actually applied in the design innovation process using our course project as an example.
- d. **Laurelle:** The most interesting thing I learned was the design methods and the different ways designers have to think through projects to get a final result.